

CBMR: Coordinate-based meta-regression for group and covariate inference

Standalone expanded mathematical appendix to Yu et al. (2025)

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Abstract

This editable appendix develops the model and inference calculations for Yu et al., *Imaging Neuroscience* 3 (2025), DOI [10.1162/IMAG.a.1057](https://doi.org/10.1162/IMAG.a.1057). It concerns group-specific spatial baselines and shared, globally constant study-covariate effects, exactly the regression setup of that article. Poisson compression, moment-matched negative-binomial aggregation, study-clustered negative-binomial integration, complete likelihood derivatives, structured covariance, and group and covariate tests are derived step by step. Exact algebra, working-distribution approximations, and asymptotic inference are distinguished throughout. The paper’s row-vector covariate notation and selected-group/covariate contrasts are retained, with new helper symbols explicitly distinguished. Qualifications address covariate information lost by aggregation, dispersion held fixed versus jointly estimated, identifiability of a totals-only model with free group scales, and the uncertainty interpretation of smoothing. These are self-contained handwritten derivations for manual review, not full formal-proof certification of the article or of this document.

Scope. The sole scientific reference is the supplied `docs/imag.a.1057.pdf`. This is an explanatory companion, not the article’s original supplementary appendix. Repository derivation notes provided algebraic provenance; no other manuscript’s model is required. All equations below are numbered independently of the paper.

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1 Paper-specific setup and correspondence

1.1 Indices, dimensions, and the regression model

Let $\mathcal{I}_g$ be the set of studies in group g , $M_g = |\mathcal{I}_g|$, and $\sum_g M_g = M$. There are N voxels in a fixed analysis mask and G disjoint groups. For study i , $g(i)$ denotes its group. The observed count at voxel j is Y_{ij} , with positive mean μ_{ij} . The paper notes that these counts are usually zero or one. Poisson and NB distributions nevertheless assign mass to larger counts: they are count-model approximations, not normalized Bernoulli models.

Symbol	Dimension	Meaning
X, x_j	$N \times P, P \times 1$	Cubic B-spline design; row j of X is $x_j^\top$
β_g	$P \times 1$	Spatial baseline coefficients for group g
Z, Z_i	$M \times R, 1 \times R$	Study design and its row i , in the paper's row-vector convention
γ	$R \times 1$	Shared, globally constant covariate coefficients
ϑ	$(GP + R) \times 1$	Stacked regression coefficients only
α_g	scalar	Original group NB overdispersion
$r'_{gj}, p'_{gj}, \alpha'_g$	scalars	Paper's matched NB shape, count probability, and effective dispersion
ν_g	scalar	Clustered gamma shape and rate, $1/\alpha_g$
H, I	$(GP + R) \times (GP + R)$	Observed and expected information

The symbol R always counts global covariates; it never denotes a shape. The paper's single-group and multi-group setups are captured by

$$\eta_{ij} = \log \mu_{ij} = x_j^\top \beta_{g(i)} + Z_i \gamma, \quad \vartheta = (\beta_1^\top, \dots, \beta_G^\top, \gamma^\top)^\top, \quad (1.1)$$

$$\mu_{gj}^X = \exp(x_j^\top \beta_g), \quad \mu_i^Z = \exp(Z_i \gamma), \quad \mu_{ij} = \mu_{g(i)j}^X \mu_i^Z. \quad (1.2)$$

The superscripts X and Z label the paper's spatial and study mean factors; they are not powers. In particular $Z_i \gamma$ is a scalar, $\nabla_\gamma \mu_i^Z = \mu_i^Z Z_i^\top$ is a column, and $\nabla_\gamma^2 \mu_i^Z = \mu_i^Z Z_i^\top Z_i$ is an $R \times R$ matrix. Transposing Z_i in the predictor would reverse the paper's row convention. Thus $\partial \eta_{ij} / \partial Z_{ir} = \gamma_r$ is constant over voxels. A spatially varying baseline is *not* a spatially varying moderator effect. No coefficient-map extension for moderators is developed here. The single-group stacked predictor is $(\mathbf{1}_M \otimes X)\beta + (Z \otimes \mathbf{1}_N)\gamma$, recovering paper Equation 2 with studies stacked before voxels.

1.2 Native paper symbols and explicitly new helper quantities

The paper uses θ for the complete parameter vector in its likelihoods, and θ_j for a different object in voxelwise inference. We keep those meanings. The regression-only vector ϑ is an additional helper, not a redefinition of θ :

$$\theta_P = \vartheta, \quad (1.3)$$

$$\theta_{NB} = (\beta_1^\top, \dots, \beta_G^\top, \alpha'_1, \dots, \alpha'_G, \gamma^\top)^\top, \quad (1.4)$$

$$\theta_{CNB} = (\beta_1^\top, \dots, \beta_G^\top, \alpha_1, \dots, \alpha_G, \gamma^\top)^\top. \quad (1.5)$$

These match the parameter lists in paper Equations 6, 9, and 11. An alternative original-dispersion coordinate for the matched NB replaces α'_g by α_g using Equation 8; its fixed-coordinate derivatives are distinguished below.

Paper symbol	Definition and use here
Y_g, Y_{gj}	Group-voxel counts: $Y_{gj} = \sum_{i \in \mathcal{I}_g} Y_{ij}$, with $Y_g \in \mathbb{R}^N$
Y_t, Y_{ti}	Study totals: $Y_{ti} = \sum_j Y_{ij}$, with $Y_t \in \mathbb{R}^M$
μ_{gj}, μ_{ti}	Means of those two totals: $\mu_{gj} = \sum_{i \in \mathcal{I}_g} \mu_{ij}$ and $\mu_{ti} = \sum_j \mu_{ij}$
μ_g^X, μ^Z	Spatial and study mean factors, not the mean of a group total or the mean of a study total
r'_{gj}, p'_{gj}	Matched NB parameters; derivation below shows $r'_{gj} = r'_g$ for the paper's separable predictor
θ_j, V_j	The S selected groups' voxelwise values and their $S \times S$ covariance, as in paper Equation 12
C, m, S	Group contrast $C \in \mathbb{R}^{m \times S}$, with m tested constraints and S involved groups
C_γ, s	Covariate contrast $C_\gamma \in \mathbb{R}^{m \times s}$ on s selected covariates, as in paper Equation 14
W_j	Signed scalar Wald statistic of paper Equation 13; its square is the corresponding chi-square statistic

The occurrence $\mu_{g(i)j}$ in paper Equation 5 denotes study i 's cell mean with its group made explicit; we write it as μ_{ij} . Without the study index, $\mu_{gj} = \mathbb{E} Y_{gj}$ is instead the *aggregate* mean in Equations 6–9. This distinction is essential. The paper's S counts selected groups, so we write the Schur complement as $\mathcal{S}$, never plain S . The helper $\nu_g = 1/\alpha_g$ denotes clustered gamma shape and rate, while λ_i remains the paper's random study multiplier. New smoothing strengths are denoted τ_g , to keep them distinct from that random multiplier. The symbols $H, I, U, \Psi, A, s_1, s_2, h_g, T_g$ and calligraphic coefficient-contrast matrices below are explicitly defined algebraic helpers; they are not additional fitted models. Lowercase y in a scalar mass $p(y)$ is an observed integer argument.

1.3 Stacking the grouped design one row at a time

Define $L_{ig} = \mathbf{1}\{g(i) = g\}$, so $L \in \mathbb{R}^{M \times G}$. Row (i, j) of $L \otimes X$ is $e_{g(i)}^\top \otimes x_j^\top$. Row (i, j) of $Z \otimes \mathbf{1}_N$ is Z_i . Therefore, with study index varying slowest,

$$\mathcal{X} = [L \otimes X \quad Z \otimes \mathbf{1}_N], \quad \eta = \mathcal{X}\vartheta, \quad \dim(\mathcal{X}) = MN \times (GP + R). \quad (1.6)$$

The X in this formula is still the paper's $N \times P$ spatial matrix; $\mathcal{X}$ is a new name for the much larger combined design, which need not be constructed numerically. For $G = 1$, $L = \mathbf{1}_M$, giving the single-group design of paper Equation 2. This verifies the coefficient ordering used in all later block matrices.

1.4 Group-specific constant effects allowed by the paper

Paper Section 1.2.2 describes both independent group fits and a single fit with group-supported covariate columns. For a covariate row $Z_i \in \mathbb{R}^{1 \times R_0}$ whose effects are all group-specific, construct

$$Z_{i,(g,r)}^{\text{sep}} = \mathbf{1}\{g(i) = g\} Z_{ir}, \quad \gamma^{\text{sep}} = (\gamma_1^\top, \dots, \gamma_G^\top)^\top.$$

Then $Z_i^{\text{sep}} \gamma^{\text{sep}} = Z_i \gamma_{g(i)}$, and

$$\eta_{ij} = x_j^\top \beta_{g(i)} + Z_i \gamma_{g(i)}.$$

For two groups, two studies per group, and a single covariate with values $a_1, \dots, a_4$, the new study design is

$$Z^{\text{sep}} = \begin{pmatrix} a_1 & 0 \\ a_2 & 0 \\ 0 & a_3 \\ 0 & a_4 \end{pmatrix}.$$

There are still scalar moderator effects constant over voxels. This construction does not introduce spatial coefficient maps. With only group-specific parameters, independent group data, and

separate group penalties, the likelihood factors in parameter blocks $\zeta_g = (\beta_g^\top, \gamma_g^\top)^\top$. Cross derivatives in ζ_g, ζ_h vanish for $g \neq h$, and the full information and covariance are block diagonal after ordering by ζ_g . This is why the two fitting strategies agree under those conditions.

For a mixture of common and group-specific covariates, write

$$\eta_{ij} = x_j^\top \beta_{g(i)} + Z_i^{\text{common}} \gamma^{\text{common}} + Z_i^{\text{specific}} \gamma_{g(i)}^{\text{specific}}.$$

Combine $(\beta_g, \gamma_g^{\text{specific}})$ into each group block. The common coefficients form the remaining border, so the later Schur formulas apply with these enlarged group blocks. Groups are not independent after estimating that shared border. Constant-within-group covariates can still be confounded with spatial intercepts; encoding them in separate columns does not by itself create identifiability.

1.5 Correspondence with the published equations

Paper location	Subject	Expansion here
Equations 1–5	Single/multiple groups, shared covariates, separable mean	Sections 1–2
Equation 6	Poisson likelihood and compression	Section 3 , including the allocation qualification
Equations 7–9	NB moment matching and aggregate likelihood	Sections 4–5
Equations 10–11	Within-study dependence and clustered likelihood	Section 6
Algorithm 1	Alternating dispersion and regression updates	Section 7
Equations 12–14	Group-map and global-covariate Wald inference	Sections 8–10
Section 2.1.4	Bootstrap and numerical caveats	Sections 11–12

Paper Section 2.1.2 explicitly excludes quasi-Poisson from its fitted pipeline, despite introductory discussion of earlier work. Accordingly there is no fourth fitted quasi-Poisson family in this appendix. The independent-cell NB below is the generative starting point for the paper’s moment-matched NB, not a claim that an additional independent-cell fitting option was evaluated in that article.

1.6 Identification and the meaning of information

Covariates are preprocessed and centered as described in the paper. Centering improves interpretation, but does not by itself prove rank. If $Xc = \mathbf{1}_N$ and $Z_id = b_{g(i)}$ for every study, then

$$\beta_g \mapsto \beta_g - tb_g c, \quad \gamma \mapsto \gamma + td \implies \eta_{ij} \text{ unchanged.} \quad (1.7)$$

An intercept in Z , a group-constant covariate, redundant spline columns, or combinations of such directions require constraints or removal. Constraints must also be respected when forming contrasts and covariance. Full rank, finite fitted means, and an interior finite optimum are separate requirements; an all-zero region can drive coefficients towards a boundary.

We write ℓ for log likelihood, $L = -\ell$ for negative log likelihood, $U = \nabla \ell$ for score, and

$$H = -\nabla^2 \ell = \nabla^2 L, \quad I = E_\vartheta H. \quad (1.8)$$

Dispersion is held fixed in regression derivatives unless stated otherwise. Under a correctly specified regular likelihood, differentiation under the expectation gives $EU = 0$ and $I =$

$\text{Var}(U)$. Those are distributional facts with regularity conditions. Inverting a matrix is exact algebra; identifying its inverse with estimator covariance is normally a first-order asymptotic approximation. A working likelihood may have exact derivatives without being the exact distribution of the data, and its inverse Hessian need not give robust sampling uncertainty.

2 Affine differentiation and group information blocks

2.1 From a scalar mean to a coefficient Hessian

Let e_g be column g of the G -dimensional identity and define

$$b_{ij} = \begin{pmatrix} e_{g(i)} \otimes x_j \\ Z_i^\top \end{pmatrix}, \quad \eta_{ij} = b_{ij}^\top \vartheta. \quad (2.1)$$

The cell-design vector b_{ij} is not the spatial row x_j . For a single cell, suppress its indices and let $\mu = e^\eta$. Direct differentiation gives

$$\partial_a \eta = b_a, \quad \partial_{ab} \eta = 0, \quad \partial_a \mu = \mu b_a, \quad \partial_{ab} \mu = \mu b_a b_b. \quad (2.2)$$

For any twice differentiable scalar function $\ell(\mu)$,

$$\partial_a \ell = \ell_\mu \mu b_a = \ell_\eta b_a, \quad (2.3)$$

$$\partial_{ab} \ell = \ell_{\mu\mu} \mu^2 b_a b_b + \ell_{\mu} \mu b_a b_b = \ell_{\eta\eta} b_a b_b. \quad (2.4)$$

The $\ell_{\mu\mu}$ term is essential: differentiating only the outer function with respect to μ is not the log-link Hessian. Equivalently, $d\ell = \ell_\eta b^\top d\vartheta$ and $d^2\ell = \ell_{\eta\eta} (b^\top d\vartheta)^2$.

2.2 Independent factors and spatial support

When cells are independent, write $s_{ij} = \partial \ell_{ij} / \partial \eta_{ij}$ and $w_{ij} = -\partial^2 \ell_{ij} / \partial \eta_{ij}^2$. Summing (2.4) yields

$$U = \sum_{i,j} b_{ij} s_{ij}, \quad H = \sum_{i,j} w_{ij} b_{ij} b_{ij}^\top. \quad (2.5)$$

Its individual blocks are

$$H_{\beta_g \beta_g} = \sum_{i \in \mathcal{I}_g} \sum_j w_{ij} x_j x_j^\top, \quad (2.6)$$

$$H_{\beta_g \beta_h} = 0 \quad (g \neq h), \quad (2.7)$$

$$H_{\beta_g \gamma} = \sum_{i \in \mathcal{I}_g} \sum_j w_{ij} x_j Z_i, \quad (2.8)$$

$$H_{\gamma \gamma} = \sum_{i,j} w_{ij} Z_i^\top Z_i. \quad (2.9)$$

The zero cross-group block follows from mutually exclusive membership, not from treating the shared γ as known. Its inverse can have nonzero cross-group blocks.

For a study likelihood coupling voxels, let B_i have rows b_{ij} , $s_i = \nabla_{\eta_i} \ell_i$, and $W_i = -\nabla_{\eta_i}^2 \ell_i$. Then $U_i = B_i^\top s_i$ and $H_i = B_i^\top W_i B_i$. The matrix W_i need not be diagonal. Yet a study involves only one β_g , so the same cross-group structural zeros remain. For aggregate NB, the factor is instead a group-voxel total; its full composite derivatives must be used.

3 Poisson likelihood and exact two-marginal compression

3.1 Probability mass, score, and curvature

For independent cells,

$$\Pr(Y = y \mid \mu) = e^{-\mu} \mu^y / y!, \quad (3.1)$$

$$\ell_{ij} = Y_{ij} \eta_{ij} - e^{\eta_{ij}} - \log(Y_{ij}!), \quad (3.2)$$

$$s_{ij} = Y_{ij} - \mu_{ij}, \quad w_{ij} = \mu_{ij}. \quad (3.3)$$

The score is the residual because $\partial_\eta e^\eta = e^\eta$; its derivative is $-\mu$ because the observed count is fixed. Consequently

$$U = \sum_{i,j} b_{ij}(Y_{ij} - \mu_{ij}), \quad H = I = \sum_{i,j} \mu_{ij} b_{ij} b_{ij}^\top. \quad (3.4)$$

For any a , $a^\top H a = \sum_{i,j} \mu_{ij} (b_{ij}^\top a)^2 \geq 0$. At finite positive means it is strictly positive for nonzero a exactly when the stacked regression design has no null direction. This proves convexity of the negative log likelihood, not existence of a finite minimizer in a data-poor design.

3.2 Compress the full likelihood without discarding covariate data

Define the two observed margins and two mean totals

$$Y_{gj} = \sum_{i \in \mathcal{I}_g} Y_{ij}, \quad Y_{ti} = \sum_j Y_{ij}, \quad h_g = \sum_j \mu_{gj}^X, \quad T_g = \sum_{i \in \mathcal{I}_g} \mu_i^Z. \quad (3.5)$$

Separate each part of the cell sum:

$$\sum_{i,j} Y_{ij} x_j^\top \beta_{g(i)} = \sum_g Y_g^\top X \beta_g, \quad (3.6)$$

$$\sum_{i,j} Y_{ij} Z_i \gamma = \sum_i Y_{ti} Z_i \gamma, \quad (3.7)$$

$$\sum_{i,j} \mu_{ij} = \sum_g \left(\sum_j \mu_{gj}^X \right) \left(\sum_{i \in \mathcal{I}_g} \mu_i^Z \right) = \sum_g h_g T_g. \quad (3.8)$$

Thus, up to a parameter-independent data term,

$$\ell_{\text{full}} = \sum_g Y_g^\top X \beta_g + \sum_i Y_{ti} Z_i \gamma - \sum_g h_g T_g + c(Y). \quad (3.9)$$

Both spatial totals Y_g and study totals Y_{ti} are retained. This is an exact compression of the *full* Poisson likelihood. The factorial data term may be dropped for parameter optimization, but likelihood comparisons requiring absolute normalizations must keep track of which observation space is being used.

3.3 Differentiate the compressed form explicitly

Define

$$a_g = X^\top \mu_g^X, \quad B_g = X^\top \text{diag}(\mu_g^X) X, \quad (3.10)$$

$$t_g^{(z)} = \sum_{i \in \mathcal{I}_g} \mu_i^Z Z_i^\top, \quad V_g = \sum_{i \in \mathcal{I}_g} \mu_i^Z Z_i^\top Z_i. \quad (3.11)$$

Then $\nabla_{\beta_g} h_g = a_g$, $\nabla_{\beta_g}^2 h_g = B_g$, $\nabla_{\gamma} T_g = t_g^{(z)}$, and $\nabla_{\gamma}^2 T_g = V_g$. Differentiating (3.9) once gives

$$U_{\beta_g} = X^\top Y_g - T_g a_g, \quad U_{\gamma} = \sum_i Y_{ti} Z_i^\top - \sum_g h_g t_g^{(z)}. \quad (3.12)$$

Differentiating again and negating gives

$$D_g = T_g B_g, \quad F_g = a_g (t_g^{(z)})^\top, \quad E = \sum_g h_g V_g. \quad (3.13)$$

Here $D_g = H_{\beta_g \beta_g}$, $F_g = H_{\beta_g \gamma}$, and $E = H_{\gamma \gamma}$; E is a matrix, whereas E is expectation. These formulas reproduce the cell sums without an $M \times N$ mean array. For example the $P \times R$ cross block comes from differentiating $-T_g a_g$ with respect to γ , not from an independence assumption.

3.4 The two expressions in paper Equation 6 need a qualification

Poisson closure says $Y_{gj} \sim \text{Poisson}(\mu_{gj}^X T_g)$. The likelihood using *only* these totals is

$$\ell_{\text{totals}} = \sum_g Y_g^\top X \beta_g + \sum_g Y_g \log T_g - \sum_g h_g T_g + c_{\text{tot}}(Y), \quad Y_g = \sum_{i \in \mathcal{I}_g} Y_{ti}. \quad (3.14)$$

This corresponds to the totals-only first expression in paper Equation 6. The paper's later expression retaining study totals corresponds to (3.9). Subtraction exposes the distinction:

$$\ell_{\text{full}} - \ell_{\text{totals}} = \sum_g \sum_{i \in \mathcal{I}_g} Y_{ti} \log(\mu_i^Z / T_g) + c(Y) - c_{\text{tot}}(Y). \quad (3.15)$$

Conditional on a voxel total, Poisson allocations across studies are multinomial with probabilities $\pi_{ig} = \mu_i^Z / T_g$. Indeed, dividing the product cell mass by the total mass gives $Y_{gj}! \prod_i \pi_{ig}^{Y_{ij}} / \prod_i Y_{ij}!$. Summing its log over voxels gives exactly (3.15).

For fixed γ , this difference is constant in β . Without estimated covariates it is also constant in the remaining regression parameters. In general it is not constant in estimated γ . Writing $\bar{z}_g = t_g^{(z)} / T_g$,

$$U_{\text{alloc},g} = \sum_{i \in \mathcal{I}_g} Y_{ti} Z_i^\top - Y_g \bar{z}_g, \quad (3.16)$$

$$H_{\text{alloc},g} = Y_g \{V_g / T_g - \bar{z}_g \bar{z}_g^\top\} = Y_g \text{Cov}_{\pi_g}(Z_i^\top) \succeq 0. \quad (3.17)$$

Therefore Poisson closure does not justify blanket equivalence of totals-only and full likelihoods when covariate effects are estimated. The Poisson limit of the aggregate NB likelihood below is the totals-only likelihood, not automatically the full two-margin expression.

4 Independent NB cells as the basis of moment matching

4.1 NB2 convention and the underlying probability mass

Within group g , put $r = 1/\alpha_g > 0$. An independent gamma multiplier with shape and rate r has mean one and variance α_g . Mixing a Poisson mean μ by that multiplier gives

$$p(y | \mu, r) = \frac{\Gamma(y+r)}{\Gamma(r)\Gamma(y+1)} \left(\frac{r}{r+\mu}\right)^r \left(\frac{\mu}{r+\mu}\right)^y, \quad \text{Var}(Y) = \mu + \alpha_g \mu^2. \quad (4.1)$$

The cells in this construction have independent multipliers, unlike the common study multiplier in Section 6. At fixed r , the scalar log likelihood is

$$\ell = \log \Gamma(y+r) - \log \Gamma(r) - \log \Gamma(y+1) + r \log r + y \eta - (y+r) \log(r + e^\eta). \quad (4.2)$$

Retain the gamma terms if dispersion is differentiated. They are constant only during the regression step at fixed dispersion.

4.2 Scalar derivatives, information, and their limitations

First combine the score numerator:

$$\ell_\eta = y - (y + r) \frac{\mu}{r + \mu} = \frac{r(y - \mu)}{r + \mu} = \frac{y - \mu}{1 + \alpha_g \mu}. \quad (4.3)$$

The quotient rule now gives

$$\ell_{\eta\eta} = \frac{-\mu(1 + \alpha_g \mu) - \alpha_g \mu(y - \mu)}{(1 + \alpha_g \mu)^2} = -\frac{\mu(1 + \alpha_g y)}{(1 + \alpha_g \mu)^2}, \quad (4.4)$$

$$w^{\text{obs}} = \frac{\mu(1 + \alpha_g y)}{(1 + \alpha_g \mu)^2}, \quad w^{\text{F}} = \mathbb{E} w^{\text{obs}} = \frac{\mu}{1 + \alpha_g \mu}. \quad (4.5)$$

Also $\text{Var}(\ell_\eta) = (\mu + \alpha_g \mu^2)/(1 + \alpha_g \mu)^2 = w^{\text{F}}$. Substitution in (2.5)–(2.9) gives all underlying independent-cell regression blocks. Their observed curvature is nonnegative at fixed α_g , and both weights tend to μ as $\alpha_g \downarrow 0$. The denominators contain cell means; simply replacing Y_{ij} by Y_{gj} does not produce the paper’s moment-matched likelihood. This cell-level derivation identifies the original dispersion convention and clarifies why Poisson compression cannot be reused unaltered.

4.3 Shape derivatives for an optional joint cell-model calculation

Define $\psi = d \log \Gamma / dr$ and $\psi_1 = d\psi / dr$. At fixed μ the exact derivatives are

$$\ell_r = \psi(y + r) - \psi(r) + \log r + 1 - \log(r + \mu) - \frac{y + r}{r + \mu}, \quad (4.6)$$

$$\ell_{rr} = \psi_1(y + r) - \psi_1(r) + \frac{1}{r} - \frac{1}{r + \mu} - \frac{\mu - y}{(r + \mu)^2}, \quad (4.7)$$

$$\ell_{\eta r} = \frac{\mu(y - \mu)}{(r + \mu)^2}. \quad (4.8)$$

Thus $H_{\vartheta r} = -\sum b_{ij} \ell_{\eta r}$ within the relevant group. Under this correctly specified cell model, $\mathbb{E} \ell_{\eta r} = 0$. Observed cross information need not vanish. Neither that orthogonality nor fixed-dispersion convexity transfers automatically to the composed aggregate likelihood with γ -dependent shape.

5 Moment-matched aggregate NB: full composite differentiation

5.1 Match the mean and variance, not the entire distribution

Temporarily omit the fixed group label g : $Y_j = Y_{gj}$, $\mu_j = \mu_{gj}$, $\mu_j^X = \mu_{gj}^X$, $\alpha = \alpha_g$, $\alpha' = \alpha'_g$, and $p'_j = p'_{gj}$. The study index i still ranges over $\mathcal{I}_g$. After deriving voxel-constant shape we write $r' = r'_g = r'_{gj}$. Let

$$s_1 = \sum_i \mu_i^Z, \quad s_2 = \sum_i (\mu_i^Z)^2, \quad s_3 = \sum_i (\mu_i^Z)^3. \quad (5.1)$$

Independence of the original NB cells gives

$$\mu_j = \mathbb{E} Y_j = \mu_j^X s_1, \quad \text{Var}(Y_j) = \mu_j^X s_1 + \alpha(\mu_j^X)^2 s_2. \quad (5.2)$$

An NB variable of mean μ_j and shape r' has variance $\mu_j + \mu_j^2/r'$. Equating extra variances and cancelling $(\mu_j^X)^2$ gives

$$r' = \frac{s_1^2}{\alpha s_2}, \quad A = \frac{s_1}{\alpha s_2} = \frac{r'}{s_1}, \quad (5.3)$$

$$p'_j = \frac{\mu_j}{\mu_j + r'} = \frac{\mu_j^X}{\mu_j^X + A}, \quad \alpha' = \frac{1}{r'} = \frac{\alpha s_2}{s_1^2}. \quad (5.4)$$

Our p'_j is the probability appearing as $p'^{Y_j}_j$ in the NB mass. These are paper Equations 7–8 specialized to its separable mean. The effective shape is constant over voxels within a group, but generally depends on γ .

Recover the paper's primed parameters explicitly. Define the extra variance $\Delta_{gj} = \alpha_g \sum_{i \in \mathcal{I}_g} \mu_{ij}^2$. Here $s_{kg} = \sum_{i \in \mathcal{I}_g} (\mu_i^Z)^k$ restores the group index suppressed in s_k . For an NB with count probability p' and shape r' , $\mu = r'p'/(1-p')$ and $\text{Var}(Y) = r'p'/(1-p')^2$. Subtracting its mean gives $\text{Var}(Y) - \mu = \mu^2/r'$. Equating this to Δ_{gj} and then solving for p' yields

$$r'_{gj} = \frac{\mu_{gj}^2}{\Delta_{gj}} = \frac{(\mu_{gj}^X s_{1g})^2}{\alpha_g (\mu_{gj}^X)^2 s_{2g}} = \frac{s_{1g}^2}{\alpha_g s_{2g}} = r'_g, \quad (5.5)$$

$$p'_{gj} = \frac{\mu_{gj}}{\mu_{gj} + r'_{gj}} = \frac{\Delta_{gj}}{\mu_{gj} + \Delta_{gj}} = \frac{\sum_{i \in \mathcal{I}_g} \mu_{ij}^2}{\alpha_g^{-1} \mu_{gj} + \sum_{i \in \mathcal{I}_g} \mu_{ij}^2}, \quad (5.6)$$

$$\alpha'_g = \frac{\Delta_{gj}}{\mu_{gj}^2} = \frac{\alpha_g s_{2g}}{s_{1g}^2} = \frac{1}{r'_g}. \quad (5.7)$$

This is a dimensionally complete derivation of the moment-matching relations in paper Equations 7–8, including the probability convention. Only the probability varies over voxels in this separable setup. The parameter $A_g = r'_g/s_{1g}$ is an additional helper introduced for differentiation, not a replacement for the paper's r'_{gj} or α'_g .

The generating function of an original cell is $(1 + \alpha \mu_j^X \mu_i^Z (1-t))^{-1/\alpha}$. Its product over studies is generally not one NB generating function. If all $\mu_i^Z = \mu^Z$, the probabilities are common and the product is $(1 + \alpha \mu_j^X \mu^Z (1-t))^{-M_g/\alpha}$: the sum is exactly NB. More generally NB closure requires a common probability parameter. Here, with common original α , equal positive μ_i^Z supplies it.

To see what matching misses, the NB cumulant generating function is $K(t) = -\alpha^{-1} \log\{1 - \alpha \mu(e^t - 1)\}$. To show all three differentiations, put $D(t) = 1 - \alpha \mu(e^t - 1)$, so $D'(t) = -\alpha \mu e^t$. The successive derivatives are

$$K'(t) = \mu e^t / D(t), \quad (5.8)$$

$$K''(t) = \mu e^t / D(t) + \alpha \mu^2 e^{2t} / D(t)^2, \quad (5.9)$$

$$K'''(t) = \mu e^t / D(t) + 3\alpha \mu^2 e^{2t} / D(t)^2 + 2\alpha^2 \mu^3 e^{3t} / D(t)^3. \quad (5.10)$$

Using $D(0) = 1$ gives $K'''(0) = \mu + 3\alpha \mu^2 + 2\alpha^2 \mu^3$. Add the true independent cumulants and subtract the matched one:

$$\kappa_3^{\text{true}} - \kappa_3^{\text{matched}} = 2\alpha^2 (\mu_j^X)^3 \left(s_3 - \frac{s_2^2}{s_1} \right) \geq 0. \quad (5.11)$$

The inequality is Cauchy–Schwarz, $(\sum_i (\mu_i^Z)^2)^2 \leq (\sum_i \mu_i^Z)(\sum_i (\mu_i^Z)^3)$. Equality for positive μ_i^Z requires equal μ_i^Z . Moment matching is a working distribution, not a general convolution identity.

5.2 Rewrite paper Equation 9 before differentiating

Set $\eta_j = x_j^\top \beta$, $\mu_j^X = e^{\eta_j}$, $d_j = \mu_j^X + A$. Ignoring only count-factorial terms, the negative log likelihood is

$$\begin{aligned} \Psi(\eta, r', A) = & - \sum_j \log \Gamma(Y_j + r') + N \log \Gamma(r') - Nr' \log A \\ & + \sum_j (r' + Y_j) \log(\mu_j^X + A) - \sum_j Y_j \eta_j. \end{aligned} \quad (5.12)$$

Indeed, $\log p'_j = \eta_j - \log d_j$ and $\log(1 - p'_j) = \log A - \log d_j$. First treat (η, r', A) as independent coordinates. Only afterwards substitute $r'(\gamma, \alpha)$ and $A(\gamma, \alpha)$. This order keeps the nonlinear inner-map curvature visible.

5.3 All outer first and second partial derivatives

At fixed r', A ,

$$\Psi_{\eta_j} = (r' + Y_j) \frac{\mu_j^X}{d_j} - Y_j, \quad (5.13)$$

$$\partial_{\eta_j}(\mu_j^X/d_j) = \frac{\mu_j^X d_j - (\mu_j^X)^2}{d_j^2} = \frac{\mu_j^X A}{d_j^2}, \quad (5.14)$$

$$\Psi_{\eta_j \eta_j} = \frac{(r' + Y_j) \mu_j^X A}{d_j^2} = w_j, \quad \Psi_{\eta_j \eta_k} = 0 \quad (j \neq k), \quad (5.15)$$

$$\Psi_{\eta_j r'} = \frac{\mu_j^X}{d_j}, \quad \Psi_{\eta_j A} = -\frac{(r' + Y_j) \mu_j^X}{d_j^2}. \quad (5.16)$$

The shape and scale derivatives are

$$\Psi_{r'} = - \sum_j \psi(Y_j + r') + N \psi(r') - N \log A + \sum_j \log d_j, \quad (5.17)$$

$$\Psi_A = -Nr'/A + \sum_j (r' + Y_j)/d_j, \quad (5.18)$$

$$\Psi_{r' r'} = - \sum_j \psi_1(Y_j + r') + N \psi_1(r'), \quad (5.19)$$

$$\Psi_{r' A} = -N/A + \sum_j d_j^{-1}, \quad (5.20)$$

$$\Psi_{AA} = Nr'/A^2 - \sum_j (r' + Y_j)/d_j^2. \quad (5.21)$$

For instance $\partial_A(-Nr'/A) = Nr'/A^2$, whereas differentiating each reciprocal d_j^{-1} contributes a minus. Together (5.16)–(5.21) specify every first and second outer derivative; omitted off-diagonal η entries are zero, and reversed mixed entries follow by symmetry.

5.4 Inner gradients and Hessians: the factors two and four

All derivatives in this subsection are with respect to γ , holding the *original* α fixed. Since $\mu_i^Z = \exp(Z_i \gamma)$,

$$\nabla \mu_i^Z = \mu_i^Z Z_i^\top, \quad \nabla^2 \mu_i^Z = \mu_i^Z Z_i^\top Z_i, \quad \nabla((\mu_i^Z)^2) = 2(\mu_i^Z)^2 Z_i^\top, \quad \nabla^2((\mu_i^Z)^2) = 4(\mu_i^Z)^2 Z_i^\top Z_i. \quad (5.22)$$

Define $b_k = \sum_i (\mu_i^Z)^k Z_i^\top$ and $C_k = \sum_i (\mu_i^Z)^k Z_i^\top Z_i$, for $k = 1, 2$. Then

$$\nabla s_1 = b_1, \quad \nabla^2 s_1 = C_1, \quad \nabla s_2 = 2b_2, \quad \nabla^2 s_2 = 4C_2. \quad (5.23)$$

Apply $\nabla^2 \log s = s^{-1} \nabla^2 s - s^{-2} \nabla s \nabla s^\top$:

$$l = \nabla \log s_1 = b_1/s_1, \quad L_1 = \nabla^2 \log s_1 = C_1/s_1 - b_1 b_1^\top / s_1^2, \quad (5.24)$$

$$h = \nabla \log s_2 = 2b_2/s_2, \quad L_2 = \nabla^2 \log s_2 = 4C_2/s_2 - 4b_2 b_2^\top / s_2^2. \quad (5.25)$$

With $q = \log r'$ and $a = \log A$,

$$q = 2 \log s_1 - \log s_2 - \log \alpha, \quad d_q = \nabla q = 2l - h, \quad Q_q = \nabla^2 q = 2L_1 - L_2, \quad (5.26)$$

$$a = \log s_1 - \log s_2 - \log \alpha, \quad d_a = \nabla a = l - h, \quad Q_a = \nabla^2 a = L_1 - L_2. \quad (5.27)$$

Finally differentiation of e^q and e^a gives

$$r_\gamma := \nabla r' = r' d_q, \quad R_\gamma := \nabla^2 r' = r' (d_q d_q^\top + Q_q), \quad (5.28)$$

$$a_\gamma := \nabla A = A d_a, \quad A_\gamma := \nabla^2 A = A (d_a d_a^\top + Q_a). \quad (5.29)$$

The capital R_γ here is a Hessian, not the scalar number R . The outer-product terms arise from differentiating the exponential prefactor and must not be discarded.

5.5 The complete regression score and group blocks

The spatial score and information are immediately

$$U_\beta = -X^\top \Psi_\eta, \quad D = X^\top \text{diag}(w) X. \quad (5.30)$$

For a global component a , write $r'_a = \partial r' / \partial \gamma_a$, $A_a = \partial A / \partial \gamma_a$, and use two component indices for the corresponding second derivatives. Then $\partial_{\gamma_a} \Psi = \Psi_{r' r'_a} + \Psi_A A_a$. Differentiating each product with respect to γ_b gives

$$\partial_{\gamma_b} (\Psi_{r' r'_a}) = \Psi_{r' r' r'_b r'_a} + \Psi_{r' A} A_b r'_a + \Psi_{r' r'_a b}, \quad (5.31)$$

$$\partial_{\gamma_b} (\Psi_A A_a) = \Psi_{r' A} r'_b A_a + \Psi_{AA} A_b A_a + \Psi_A A_{ab}. \quad (5.32)$$

Thus the group contribution to the global score and Hessian is

$$U_{\gamma, g} = -\Psi_{r' r_\gamma} - \Psi_A a_\gamma, \quad (5.33)$$

$$E_g = \Psi_{r' r' r_\gamma r_\gamma^\top} + \Psi_{r' A} (r_\gamma a_\gamma^\top + a_\gamma r_\gamma^\top) + \Psi_{AA} a_\gamma a_\gamma^\top + \Psi_{r' R_\gamma} + \Psi_A A_\gamma. \quad (5.34)$$

The mixed spatial/global block is

$$F_g = \sum_j x_j \left\{ \frac{\mu_j^X}{d_j} r_\gamma^\top - \frac{(r' + Y_j) \mu_j^X}{d_j^2} a_\gamma^\top \right\}. \quad (5.35)$$

Restore the group subscripts on all inner quantities, set $D = \text{blkdiag}(D_1, \dots, D_G)$, stack F_g , and sum $E = \sum_g E_g$. Likewise $U_\gamma = \sum_g U_{\gamma, g}$. This accounts for every regression derivative at fixed original $\alpha_1, \dots, \alpha_G$.

In particular $\Psi_{r' R_\gamma} + \Psi_A A_\gamma$ does not vanish merely because the composite score is zero. An optimum imposes the available parameter score equations, not independent optimization over unconstrained r' and A for every group. Discarding these terms is a curvature approximation, not this Hessian.

5.6 Holding effective dispersion fixed is a different operation

Paper Equations 8–9 introduce effective dispersion α' . Its use as an optimization coordinate must be distinguished from holding original α fixed in (5.27). If α' is held fixed, then

$$q = -\log \alpha', \quad a = q - \log s_1, \quad d_q = 0, \quad Q_q = 0, \quad d_a = -l, \quad Q_a = -L_1. \quad (5.36)$$

These derivatives differ from (5.27). The matched mean is still $\mu_j^X s_1$, but the shape no longer changes with γ during that partial update. Because $\alpha' = \alpha s_2 / s_1^2$, a joint, unconstrained positive dispersion parameterization can describe the same matched distributions. However, a fixed-dispersion regression slice, an alternating update path, and a conditional information block depend on which coordinate is held. Joint covariance of the common regression coordinates is invariant under a smooth nuisance reparameterization when transformed consistently on identified coordinates where that covariance exists.

Direct Hessian in the paper's effective-dispersion coordinate. At fixed α' , the aggregate log mean is $\tilde{\eta}_j = x_j^\top \beta + \log s_1$, not an affine function of γ . Define

$$\tilde{s}_j = \frac{Y_j - \mu_j}{1 + \alpha' \mu_j}, \quad w_j = \frac{\mu_j(1 + \alpha' Y_j)}{(1 + \alpha' \mu_j)^2}.$$

The tilde distinguishes these scores from the study moment sums s_1, s_2 ; write $\tilde{s} = (\tilde{s}_j)_j$ and $w = (w_j)_j$ below. The derivatives of the predictor are $\nabla_\beta \tilde{\eta}_j = x_j$, $\nabla_\gamma \tilde{\eta}_j = l$, and $\nabla_\gamma^2 \tilde{\eta}_j = L_1$. Applying the full composite rule gives the group contributions

$$U_\beta = X^\top \tilde{s}, \quad U_{\gamma,g} = l \sum_j \tilde{s}_j, \quad (5.37)$$

$$D_g = X^\top \text{diag}(w) X, \quad F_g = (X^\top w) l^\top, \quad (5.38)$$

$$E_g = \left(\sum_j w_j \right) l l^\top - \left(\sum_j \tilde{s}_j \right) L_1. \quad (5.39)$$

In the last line, $-\sum_j \tilde{s}_j L_1$ comes from $-\ell_{\tilde{\eta}_j} \nabla_\gamma^2 \tilde{\eta}_j$. Dropping it away from a zero spatial-intercept score is not the observed Hessian. These formulas are the direct reduction of the outer/inner calculation with (5.36).

5.7 An identification consequence of using only matched totals

Suppose the spatial basis contains a freely fitted intercept $X_{CB} = \mathbf{1}_N$. For any two covariate vectors γ and $\tilde{\gamma}$, set

$$\tilde{\beta}_g = \beta_g + \log \left\{ \frac{s_{1g}(\gamma)}{s_{1g}(\tilde{\gamma})} \right\} c_B, \quad \tilde{\alpha}'_g = \alpha'_g.$$

Then

$$\tilde{\mu}_{gj} = \exp(x_j^\top \tilde{\beta}_g) s_{1g}(\tilde{\gamma}) = \exp(x_j^\top \beta_g) s_{1g}(\gamma) = \mu_{gj}, \quad (5.40)$$

$$\tilde{r}'_{gj} = 1/\tilde{\alpha}'_g = r'_{gj}. \quad (5.41)$$

Every factor in the matched-totals likelihood is unchanged. Consequently, these totals alone do not identify the shared γ separately from freely fitted group intercepts, even when the study design has full rank and varies within groups. This invariance holds with effective dispersion fixed or freely estimated. With original dispersion freely estimated, the same invariance uses

$$\tilde{\alpha}_g = \alpha'_g \frac{s_{1g}(\tilde{\gamma})^2}{s_{2g}(\tilde{\gamma})}.$$

It need not preserve a *fixed original* α_g ; shape variation can then carry additional information.

The singular curvature is also visible directly. In (5.39), $D_g c_B = X^\top w$, so $F_g = D_g c_B l^\top$. At a spatial-intercept stationary point, $0 = c_B^\top U_\beta = \sum_j \tilde{s}_j$ and hence

$$E_g = \left(\sum_j w_j \right) l l^\top = F_g^\top D_g^{-1} F_g.$$

Summing over groups gives $\mathcal{S} = 0$ for the covariate border when each D_g is invertible. Under the matched law the same cancellation holds for expected information because $E \tilde{s}_j = 0$. Thus a positive-definite full regression inverse cannot be assumed for this unrestricted totals-only setup.

This is a consequence of the observation model written in paper Equation 9, not an assertion about an implementation that also retains study-level allocation information. The full two-margin Poisson and full clustered likelihoods retain that information; the totals-only matched NB likelihood does not. Scientifically specified restrictions or additional data can change identifiability. A numerical ridge, or a roughness penalty that leaves constant maps unpenalized, does not establish identification from the original totals likelihood.

5.8 Expected information under the matched distribution

If the matched NB law is itself assumed, the outer scores have mean zero. In particular $E \Psi_{r'} = E \Psi_A = 0$, so inner-map curvature terms drop *in expectation*. At $\mu_j = r' \mu_j^X / A$ the necessary expectations are

$$E w_j = (r' + \mu_j) \mu_j^X A / d_j^2 = r' \mu_j^X / d_j, \quad (5.42)$$

$$E \Psi_{r'r'} = N \psi_1(r') - \sum_j E \{ \psi_1(Y_j + r') \}, \quad (5.43)$$

$$E \Psi_{r'A} = -N/A + \sum_j d_j^{-1}, \quad (5.44)$$

$$E \Psi_{AA} = N r' / A^2 - \sum_j (r' + \mu_j) / d_j^2, \quad (5.45)$$

$$E \Psi_{\eta_j r'} = \mu_j^X / d_j, \quad E \Psi_{\eta_j A} = -(r' + \mu_j) \mu_j^X / d_j^2. \quad (5.46)$$

Insert these into (5.34)–(5.35). The trigamma expectation is an NB-weighted sum over nonnegative counts; it cannot generally be replaced by $\psi_1(\mu_j + r')$.

Under the *original* unequal-probability independent-cell sums, the first two moments agree, but the expected digamma in the shape score need not agree. Thus the matched shape score need not be centered under the original distribution. Since r' depends on γ at fixed original dispersion, the matched global regression score need not be centered at the original parameter either. An exact working Hessian does not resolve this possible change of target. A robust variance, when justified, quantifies a working estimator's variability; it does not automatically remove approximation bias.

5.9 Stable log coordinates and exact recurrence checks

Let $t_j = \eta_j - a$, $p'_j = (1 + e^{-t_j})^{-1}$, $\bar{p}'_j = 1 - p'_j$. Then (5.12) becomes

$$\Psi = - \sum_j \log \Gamma(Y_j + r') + N \log \Gamma(r') + \sum_j (r' + Y_j) \text{softplus}(t_j) - \sum_j Y_j t_j. \quad (5.47)$$

Use $\text{softplus}(t) = \max(t, 0) + \log(1 + e^{-|t|})$ for evaluation. In coordinates (η, q, a) , direct differentiation gives

$$\Psi_a = \sum_j \{Y_j - (r' + Y_j)p'_j\}, \quad (5.48)$$

$$\Psi_{aa} = \sum_j w_j, \quad w_j = (r' + Y_j)p'_j \bar{p}'_j, \quad (5.49)$$

$$\Psi_q = r' \left\{ \sum_j \text{softplus}(t_j) - \sum_j [\psi(Y_j + r') - \psi(r')] \right\}, \quad (5.50)$$

$$\Psi_{qa} = -r' \sum_j p'_j, \quad (5.51)$$

$$\Psi_{qq} = \Psi_q + r'^2 \sum_j \{\psi_1(r') - \psi_1(Y_j + r')\}, \quad (5.52)$$

$$\Psi_{\eta_j q} = r' p'_j, \quad \Psi_{\eta_j a} = -w_j. \quad (5.53)$$

In particular $\partial_q(r' \Psi_{r'}) = \Psi_q + r'^2 \Psi_{r' r'}$: the first derivative appears again through reparameterization. An equivalent form of the regression blocks is

$$F_g = X^\top (r' p') d_q^\top - X^\top w d_a^\top, \quad (5.54)$$

$$E_g = \Psi_{qq} d_q d_q^\top + \Psi_{qa} (d_q d_a^\top + d_a d_q^\top) + \Psi_{aa} d_a d_a^\top + \Psi_q Q_q + \Psi_a Q_a. \quad (5.55)$$

This is a rearrangement, not a smaller-term Hessian approximation. For integer $y \geq 0$, gamma recurrence yields exact finite sums

$$\log \Gamma(r' + y) - \log \Gamma(r') = \sum_{k=0}^{y-1} \log(r' + k), \quad (5.56)$$

$$\psi(r' + y) - \psi(r') = \sum_{k=0}^{y-1} (r' + k)^{-1}, \quad (5.57)$$

$$\psi_1(r') - \psi_1(r' + y) = \sum_{k=0}^{y-1} (r' + k)^{-2}. \quad (5.58)$$

Empty sums are zero. These provide manual checks and can avoid subtracting nearby special-function values. Log-sum-exp evaluation of $\log s_1$ and $\log s_2$ likewise avoids overflowing the μ_i^Z .

5.10 Explicit optional log-dispersion blocks

For a joint matched-model calculation let $\delta = \log \alpha$ in one group. Then $q_\delta = a_\delta = -1$ and $q_{\gamma\delta} = a_{\gamma\delta} = q_{\delta\delta} = a_{\delta\delta} = 0$. The complete additional negative-log-likelihood derivatives are

$$\Psi_\delta = -\Psi_q - \Psi_a, \quad (5.59)$$

$$H_{\delta\delta} = \Psi_{qq} + 2\Psi_{qa} + \Psi_{aa}, \quad (5.60)$$

$$H_{\gamma\delta} = -(\Psi_{qq} + \Psi_{qa})d_q - (\Psi_{qa} + \Psi_{aa})d_a, \quad (5.61)$$

$$H_{\beta\delta} = -X^\top (r' p') + X^\top w. \quad (5.62)$$

Different group dispersions have no direct cross derivatives. The regression blocks remain (5.55), including their inner curvature terms. For effective log dispersion $\delta' = \log \alpha'$, the same outer formulas hold but with the inner derivatives (5.36). These formulas state both joint parameterizations without silently changing what “fixed dispersion” means.

6 Clustered NB: integrate the shared study multiplier

6.1 Gamma integral and full covariance

For $i \in \mathcal{I}_g$, put $\nu = \nu_g = 1/\alpha_g$ and assume

$$\lambda_i \sim \text{Gamma}(\text{shape } \nu, \text{rate } \nu), \quad Y_{ij} \mid \lambda_i \text{ independent Poisson}(\lambda_i \mu_{ij}). \quad (6.1)$$

Different studies have independent multipliers. Let $Y_{ti} = \sum_j Y_{ij}$ and $\mu_{ti} = \sum_j \mu_{ij} = \mu_i^Z h_g$. Multiply the conditional mass by the gamma density and integrate:

$$p(Y_i) = \frac{\nu^\nu \prod_j \mu_{ij}^{Y_{ij}}}{\Gamma(\nu) \prod_j Y_{ij}!} \int_0^\infty \lambda^{Y_{ti} + \nu - 1} e^{-(\mu_{ti} + \nu)\lambda} d\lambda \quad (6.2)$$

$$= \frac{\Gamma(Y_{ti} + \nu)}{\Gamma(\nu)} \frac{\nu^\nu}{(\mu_{ti} + \nu)^{Y_{ti} + \nu}} \prod_j \frac{\mu_{ij}^{Y_{ij}}}{Y_{ij}!}. \quad (6.3)$$

The gamma integral is $\int_0^\infty x^{a-1} e^{-bx} dx = \Gamma(a)b^{-a}$, $a, b > 0$. Its log is

$$\ell_i = \log \Gamma(Y_{ti} + \nu) - \log \Gamma(\nu) + \nu \log \nu - (Y_{ti} + \nu) \log(\mu_{ti} + \nu) + \sum_j Y_{ij} \eta_{ij} - \sum_j \log(Y_{ij}!). \quad (6.4)$$

Summing (6.4) and using both data margins recovers paper Equation 11 up to data-only factorials.

The law of total covariance gives, including diagonal entries,

$$\text{Cov}(Y_{ij}, Y_{ik}) = \text{E}\{\text{Cov}(Y_{ij}, Y_{ik} \mid \lambda_i)\} + \text{Cov}(\lambda_i \mu_{ij}, \lambda_i \mu_{ik}) \quad (6.5)$$

$$= \mathbf{1}_{\{j=k\}} \mu_{ij} + \alpha_g \mu_{ij} \mu_{ik}, \quad (6.6)$$

$$\text{Var}(Y_i) = \text{diag}(\mu_i) + \alpha_g \mu_i \mu_i^\top. \quad (6.7)$$

For distinct voxels this reduces to paper Equation 10. Different studies have zero covariance under the model. The diagonal is $\mu_{ij} + \alpha_g \mu_{ij}^2$, not just $\alpha_g \mu_{ij}^2$. Also $\text{Var}(Y_{ti}) = \mu_{ti} + \alpha_g \mu_{ti}^2$. This construction captures shared study fluctuation; it does not introduce an additional independent voxel-level overdispersion component.

6.2 Exact factorization into a study total and spatial allocation

Conditional on λ_i , the total has law $Y_{ti} \mid \lambda_i \sim \text{Poisson}(\lambda_i \mu_{ti})$. The same gamma integral, now applied to this single count, gives

$$p(Y_{ti}) = \frac{\Gamma(Y_{ti} + \nu)}{\Gamma(\nu) Y_{ti}!} \frac{\nu^\nu \mu_{ti}^{Y_{ti}}}{(\mu_{ti} + \nu)^{Y_{ti} + \nu}}.$$

Thus Y_{ti} is exactly NB with mean μ_{ti} and shape ν . For a fixed total the independent conditional Poisson counts have multinomial probabilities

$$\pi_{ij} = \frac{\lambda_i \mu_{ij}}{\lambda_i \mu_{ti}} = \frac{\mu_{ij}}{\mu_{ti}} = \frac{\mu_{gj}^X}{h_g}, \quad g = g(i).$$

The multiplier cancels, so integrating it does not change this conditional multinomial distribution. Multiplying the two masses,

$$\begin{aligned} p(Y_i) &= p(Y_{ti}) \frac{Y_{ti}!}{\prod_j Y_{ij}!} \prod_j \left(\frac{\mu_{ij}}{\mu_{ti}} \right)^{Y_{ij}} \\ &= \frac{\Gamma(Y_{ti} + \nu)}{\Gamma(\nu)} \frac{\nu^\nu}{(\mu_{ti} + \nu)^{Y_{ti} + \nu}} \prod_j \frac{\mu_{ij}^{Y_{ij}}}{Y_{ij}!}. \end{aligned} \quad (6.8)$$

The factorial $Y_{ti}!$ and the power $\mu_{ti}^{Y_{ti}-\sum_j Y_{ij}} = 1$ cancel explicitly, reproducing (6.3). Unlike the independent-cell sum in Section 5, this factorization uses no moment approximation.

It also separates the sources of regression information. Let $a_g = X^\top \mu_g^X$ as above. Then

$$\nabla_{\beta_g} \log \mu_{ti} = a_g/h_g, \quad \nabla_\gamma \log \mu_{ti} = Z_i^\top.$$

The NB total score on its log-mean scale is $u_{ti} = (Y_{ti} - \mu_{ti})/(1 + \alpha_g \mu_{ti})$. The conditional allocation log likelihood, apart from data terms, is $Y_i^\top X \beta_g - Y_{ti} \log h_g$. Its spatial score is $X^\top Y_i - Y_{ti} a_g/h_g$, and its γ score is zero. Combining total and allocation scores gives

$$U_{\beta_g, i} = X^\top Y_i + (u_{ti} - Y_{ti}) a_g/h_g = X^\top Y_i - c_i \mu_i^Z a_g, \quad (6.9)$$

$$U_{\gamma, i} = u_{ti} Z_i^\top = (Y_{ti} - c_i \mu_{ti}) Z_i^\top, \quad c_i = \frac{Y_{ti} + \nu_g}{\mu_{ti} + \nu_g}. \quad (6.10)$$

The cancellation follows from $u_{ti} - Y_{ti} = -\mu_{ti}(Y_{ti} + \nu_g)/(\mu_{ti} + \nu_g)$. The total captures the constant study effect, while conditional allocation captures the normalized spatial shape.

6.3 Study score and observed information on the log-mean scale

At fixed ν , define $c_i = (Y_{ti} + \nu)/(\mu_{ti} + \nu)$. Since $\partial \mu_{ti}/\partial \eta_{ij} = \mu_{ij}$,

$$s_{ij} = \frac{\partial \ell_i}{\partial \eta_{ij}} = Y_{ij} - c_i \mu_{ij}. \quad (6.11)$$

For any voxel k , $\partial c_i/\partial \eta_{ik} = -c_i \mu_{ik}/(\mu_{ti} + \nu)$. The product rule therefore gives

$$-\frac{\partial s_{ij}}{\partial \eta_{ik}} = c_i \mathbf{1}_{\{j=k\}} \mu_{ij} - \frac{c_i \mu_{ij} \mu_{ik}}{\mu_{ti} + \nu}, \quad (6.12)$$

$$W_i^{\text{obs}} = c_i \left\{ \text{diag}(\mu_i) - \frac{\mu_i \mu_i^\top}{\mu_{ti} + \nu} \right\}. \quad (6.13)$$

The second term is a rank-one correction, not a diagonal NB weight. For $a \neq 0$, weighted Cauchy–Schwarz gives $(\sum_j \mu_{ij} a_j)^2 \leq \mu_{ti} \sum_j \mu_{ij} a_j^2$, hence

$$a^\top W_i^{\text{obs}} a \geq c_i \frac{\nu}{\mu_{ti} + \nu} \sum_j \mu_{ij} a_j^2 > 0. \quad (6.14)$$

Thus the study log-mean curvature is positive definite at finite positive means and $\nu > 0$. Projection can still lose rank through the design. Since $E Y_{ti} = \mu_{ti}$, $E c_i = 1$, giving

$$W_i^F = \text{diag}(\mu_i) - \frac{\mu_i \mu_i^\top}{\mu_{ti} + \nu}. \quad (6.15)$$

These weights and scores tend to the Poisson ones as $\nu \rightarrow \infty$. The positive curvature establishes fixed-dispersion convexity only.

6.4 Project to the paper's group and global coefficients

Use h_g , $a_g = X^\top \mu_g^X$, and $B_g = X^\top \text{diag}(\mu_g^X) X$ from the Poisson calculation. The study's nonlinear negative-log term is

$$f_i(t) = (Y_{ti} + \nu_g) \log(t + \nu_g), \quad f_i'(t) = c_i, \quad f_i''(t) = -\frac{c_i}{t + \nu_g}. \quad (6.16)$$

The total $\mu_{ti} = \mu_i^Z h_g$ satisfies

$$\nabla_{\beta_g} \mu_{ti} = \mu_i^Z a_g, \quad \nabla_{\beta_g}^2 \mu_{ti} = \mu_i^Z B_g, \quad (6.17)$$

$$\nabla_{\gamma} \mu_{ti} = \mu_{ti} Z_i^\top, \quad \nabla_{\gamma}^2 \mu_{ti} = \mu_{ti} Z_i^\top Z_i, \quad \nabla_{\beta_g \gamma}^2 \mu_{ti} = \mu_i^Z a_g Z_i. \quad (6.18)$$

The elementary composite rule is $\nabla^2 f(t) = f'(t) \nabla^2 t + f''(t) \nabla t \nabla t^\top$. If f is instead viewed as a function of h_g , its spatial block is exactly $f'_h B_g + f''_h a_g a_g^\top$, with $f'_h = c_i \mu_i^Z$ and $f''_h = -c_i (\mu_i^Z)^2 / (\mu_{ti} + \nu_g)$. Thus the full study contributions are

$$D_{g,i} = c_i \mu_i^Z B_g - \frac{c_i (\mu_i^Z)^2}{\mu_{ti} + \nu_g} a_g a_g^\top, \quad (6.19)$$

$$F_{g,i} = c_i \mu_i^Z a_g Z_i - \frac{c_i \mu_i^Z \mu_{ti}}{\mu_{ti} + \nu_g} a_g Z_i = \frac{c_i \mu_i^Z \nu_g}{\mu_{ti} + \nu_g} a_g Z_i, \quad (6.20)$$

$$E_i = c_i \mu_{ti} Z_i^\top Z_i - \frac{c_i \mu_{ti}^2}{\mu_{ti} + \nu_g} Z_i^\top Z_i = \frac{c_i \mu_{ti} \nu_g}{\mu_{ti} + \nu_g} Z_i^\top Z_i. \quad (6.21)$$

The scores, retaining the data terms, are

$$U_{\beta_g,i} = X^\top Y_i - c_i \mu_i^Z a_g, \quad U_{\gamma,i} = (Y_{ti} - c_i \mu_{ti}) Z_i^\top. \quad (6.22)$$

Sum $D_{g,i}, F_{g,i}$ over studies in group g and E_i over all studies. Replace c_i by one for expected information. No spatially varying moderator design has been introduced. The two data margins suffice for the total likelihood and derivatives at fixed dispersion; a study-level robust calculation also needs the individual study projections $X^\top Y_i$.

6.5 Cluster shape and log-dispersion derivatives

Differentiate (6.4) at fixed μ_{ti} :

$$\ell_{i,\nu} = \psi(Y_{ti} + \nu) - \psi(\nu) + \log \nu + 1 - \log(\mu_{ti} + \nu) - \frac{Y_{ti} + \nu}{\mu_{ti} + \nu}, \quad (6.23)$$

$$\ell_{i,\nu\nu} = \psi_1(Y_{ti} + \nu) - \psi_1(\nu) + \frac{1}{\nu} - \frac{1}{\mu_{ti} + \nu} - \frac{\mu_{ti} - Y_{ti}}{(\mu_{ti} + \nu)^2}, \quad (6.24)$$

$$\ell_{i,\eta_{ij}\nu} = \frac{\mu_{ij}(Y_{ti} - \mu_{ti})}{(\mu_{ti} + \nu)^2}. \quad (6.25)$$

The last derivative follows from differentiating $-c_i \mu_{ij}$. For $\delta_g = \log \alpha_g$ and $\nu_g = e^{-\delta_g}$,

$$\ell_{\delta_g} = -\nu_g \ell_{\nu_g}, \quad \ell_{\delta_g \delta_g} = \nu_g^2 \ell_{\nu_g \nu_g} + \nu_g \ell_{\nu_g}, \quad (6.26)$$

$$\ell_{\vartheta \delta_g} = -\nu_g \ell_{\vartheta \nu_g}. \quad (6.27)$$

Study derivatives are summed within the group before joint blocks are formed. At a regular group shape optimum $\ell_{\nu_g} = 0$, but the extra score term in (6.27) cannot be omitted away from it. Expected shape/regression cross information is zero for the correctly specified clustered model because $E(Y_{ti} - \mu_{ti}) = 0$. That fact need not hold for a misspecified distribution or observed Hessian.

7 Optimization, fixed slices, and nuisance adjustment

7.1 Interpret Algorithm 1 as joint estimation by alternating updates

The paper initializes the overdispersed fits from a Poisson solution and alternates dispersion updates with regression updates using L-BFGS. Mathematically, for a negative log likelihood $L(\vartheta, \delta)$, one cycle takes the schematic form

$$\delta^{(k+1)} \approx \arg \min_{\delta} L(\vartheta^{(k)}, \delta), \quad \vartheta^{(k+1)} \approx \arg \min_{\vartheta} L(\vartheta, \delta^{(k+1)}). \quad (7.1)$$

Separate subproblems do not mean the final dispersion was externally known. Check the full score, objective progress, and sensitivity to initialization. Small objective changes alone can reflect a stalled or poorly scaled fit. L-BFGS's limited-memory curvature approximation is an optimization device, not automatically the observed-information covariance used for inference.

Poisson, underlying independent NB, and clustered NB have convex negative log likelihood in the affine regression coefficients at fixed dispersion, as proved above. This does not imply joint convexity in regression and dispersion, nor convexity of the matched NB composition at fixed original dispersion. It also does not guarantee numerical convergence, full rank, or a finite estimate. These distinctions refine, rather than replace, the paper's caution about fitting its overdispersed models.

7.2 Joint versus fixed-dispersion regression covariance

Let $\delta = (\log \alpha_1, \dots, \log \alpha_G)^\top$ and partition a joint observed or expected information matrix in auxiliary coordinate order (ϑ, δ) as

$$\mathcal{H} = \begin{pmatrix} H_{\vartheta\vartheta} & \mathcal{B} \\ \mathcal{B}^\top & A_\delta \end{pmatrix}. \quad (7.2)$$

This is a reordering and log transformation of the relevant full paper parameter vector θ , not a new meaning for that vector. Use $\log \alpha'_g$ instead if effective dispersion is the coordinate being held fixed, and form all blocks in that same coordinate system. Assume the indicated blocks are invertible and the full matrix is positive definite on the identified coordinates. For a right-hand side $(b, 0)$ the nuisance equation is $\mathcal{B}^\top x_\vartheta + A_\delta x_\delta = 0$, so $x_\delta = -A_\delta^{-1} \mathcal{B}^\top x_\vartheta$. Substitute into the regression equation:

$$(H_{\vartheta\vartheta} - \mathcal{B} A_\delta^{-1} \mathcal{B}^\top) x_\vartheta = b. \quad (7.3)$$

Therefore

$$[\mathcal{H}^{-1}]_{\vartheta\vartheta} = (H_{\vartheta\vartheta} - \mathcal{B} A_\delta^{-1} \mathcal{B}^\top)^{-1}, \quad V_{\vartheta|\delta} \approx H_{\vartheta\vartheta}^{-1}. \quad (7.4)$$

The second expression treats dispersion as fixed for inference. The first allows local nuisance estimation uncertainty. With positive-definite blocks it is no smaller in the PSD ordering. Expected orthogonality can make $\mathcal{B} = 0$ in particular correctly specified models, but does not make an observed $\mathcal{B}$ identically zero. The aggregate model needs its own derivatives, not a borrowed orthogonality.

For any smooth parameter transformation $\xi = \xi(\zeta)$,

$$\nabla_\zeta^2 L = J_\xi^\top (\nabla_\xi^2 L) J_\xi + \sum_k L_{\xi_k} \nabla_\zeta^2 \xi_k. \quad (7.5)$$

This explains both the log-dispersion extra score term and the aggregate inner curvature. At $\alpha = 0$, log dispersion is at infinity; ordinary interior joint-Wald asymptotics do not address that boundary. All inverse statements require the appropriate regular parameter space.

8 Group blocks, Schur solves, and complete regression covariance

8.1 Structural zeros do not imply independent fitted groups

For each of the paper's likelihoods, with group-specific dispersion fixed, the regression information has the structure

$$H = \begin{pmatrix} D & F \\ F^\top & E \end{pmatrix}, \quad D = \text{blkdiag}(D_1, \dots, D_G), \quad F = \begin{pmatrix} F_1 \\ \vdots \\ F_G \end{pmatrix}. \quad (8.1)$$

Here D_g is $P \times P$, F_g is $P \times R$, and E is $R \times R$. The expressions are (3.13), (5.34)–(5.35), or (6.21), for the chosen family. Observed and expected versions must not be mixed without stating why. Joint nuisance blocks can be retained in a larger border or removed by the exact adjustment (7.4). The unrestricted matched-totals setup in Section 5.7 has a singular covariate border; the invertible formulas below cannot be applied to that case without additional scientifically justified identifying restrictions.

When $R = 0$, the inverse is $\text{blkdiag}(D_g^{-1})$. For $R > 0$, common γ connects groups through F . The appropriate covariance is not obtained by dropping those connections. Independently fitting each group's own γ_g would instead define a different model and, for independent datasets, independent fitted groups. It is not the joint shared-covariate fit with a block-diagonal inverse.

8.2 Derive elimination from the two block equations

Solve $Hx = b$ with $x = (x_D, x_\gamma)$ and $b = (b_D, b_\gamma)$. The equations are

$$Dx_D + Fx_\gamma = b_D, \quad (8.2)$$

$$F^\top x_D + Ex_\gamma = b_\gamma. \quad (8.3)$$

Assume D is nonsingular. Solve the first and substitute:

$$x_D = D^{-1}b_D - D^{-1}Fx_\gamma, \quad (8.4)$$

$$(E - F^\top D^{-1}F)x_\gamma = b_\gamma - F^\top D^{-1}b_D. \quad (8.5)$$

Define $\mathcal{T} = D^{-1}F$ and the Schur complement $\mathcal{S} = E - F^\top \mathcal{T}$. The calligraphic $\mathcal{T}_g = D_g^{-1}F_g$ is a $P \times R$ matrix, distinct from the scalar $T_g = \sum_{i \in \mathcal{I}_g} \mu_i^Z$. With $\mathcal{S}$ nonsingular, the algorithm is

$$y = D^{-1}b_D, \quad x_\gamma = \mathcal{S}^{-1}(b_\gamma - F^\top y), \quad x_D = y - \mathcal{T}x_\gamma. \quad (8.6)$$

Each occurrence of an inverse multiplying a vector or matrix means a linear solve with a factorization, not a demand to construct a dense inverse. The block structure allows the group solves to be performed separately.

8.3 Read off the inverse and prove definiteness conditions

The coefficients of the right-hand sides in (8.6) give

$$H^{-1} = \begin{pmatrix} D^{-1} + \mathcal{T}\mathcal{S}^{-1}\mathcal{T}^\top & -\mathcal{T}\mathcal{S}^{-1} \\ -\mathcal{S}^{-1}\mathcal{T}^\top & \mathcal{S}^{-1} \end{pmatrix}. \quad (8.7)$$

Multiplication verifies the identity. For example the upper-right block of HH^{-1} is $-D\mathcal{T}\mathcal{S}^{-1} + F\mathcal{S}^{-1} = 0$, since $D\mathcal{T} = F$. The lower-right block is $-F^\top \mathcal{T}\mathcal{S}^{-1} + E\mathcal{S}^{-1} = \mathcal{S}\mathcal{S}^{-1} = I_R$. The remaining blocks similarly give I_{GP} and zero.

For vectors a, b , complete the square:

$$\begin{pmatrix} a \\ b \end{pmatrix}^\top H \begin{pmatrix} a \\ b \end{pmatrix} = (a + \mathcal{T}b)^\top D(a + \mathcal{T}b) + b^\top \mathcal{S}b. \quad (8.8)$$

Thus with $D \succ 0$, $H \succ 0$ if and only if $\mathcal{S} \succ 0$. An invertible but indefinite matrix also has an algebraic inverse, but that inverse is not a valid likelihood covariance. For regular likelihood inference the covariance approximation is $V \approx H(\hat{\vartheta})^{-1}$ or $I(\hat{\vartheta})^{-1}$, subject to the model, asymptotic regime, and nuisance convention.

8.4 Shared covariates induce a low-rank group adjustment

Write $\mathcal{T}_g = D_g^{-1}F_g$. Equation (8.7) gives

$$V_{\beta_g\beta_h} = \mathbf{1}_{\{g=h\}}D_g^{-1} + \mathcal{T}_g\mathcal{S}^{-1}\mathcal{T}_h^\top, \quad (8.9)$$

$$V_{\beta_g\gamma} = -\mathcal{T}_g\mathcal{S}^{-1}, \quad V_{\gamma\gamma} = \mathcal{S}^{-1}. \quad (8.10)$$

The spatial adjustment $\mathcal{T}\mathcal{S}^{-1}\mathcal{T}^\top$ is PSD and has rank at most R when $\mathcal{S} \succ 0$. Its individual cross-group entries need not be positive. For a two-group spatial coefficient contrast,

$$\text{Var}(\hat{\beta}_g - \hat{\beta}_h) \approx D_g^{-1} + D_h^{-1} + (\mathcal{T}_g - \mathcal{T}_h)\mathcal{S}^{-1}(\mathcal{T}_g - \mathcal{T}_h)^\top. \quad (8.11)$$

Using the two marginal variances but dropping their covariance is wrong; using only $D_g^{-1} + D_h^{-1}$ omits shared- nuisance uncertainty unless the adjustment cancels. This cancellation can occur for a contrast even when each individual map has substantial nuisance-induced uncertainty.

8.5 Selected contrasts and marginal diagonals without dense inversion

For an auxiliary $k \times (GP + R)$ coefficient contrast $\mathcal{C}$, solve $HB = \mathcal{C}^\top$ by (8.6). Then

$$CVC^\top \approx CB. \quad (8.12)$$

For one contrast column c , solve $Hw_c = c$ and use $c^\top w_c$. For selected coefficient variances set c to the corresponding unit vectors; for map variances set it to the spline-evaluation vectors. One may batch right-hand sides in memory-limited chunks.

Partition $\mathcal{C} = (\mathcal{C}_D, \mathcal{C}_\gamma)$ and insert (8.7). These are coefficient-space blocks, not the paper's selected-group matrix C or selected-covariate matrix C_γ :

$$\mathcal{C}H^{-1}\mathcal{C}^\top = \mathcal{C}_D D^{-1} \mathcal{C}_D^\top + (\mathcal{C}_\gamma - \mathcal{C}_D \mathcal{T})\mathcal{S}^{-1}(\mathcal{C}_\gamma - \mathcal{C}_D \mathcal{T})^\top. \quad (8.13)$$

This directly displays nuisance cancellation. For group g at voxel j ,

$$\text{Var}(x_j^\top \hat{\beta}_g) \approx x_j^\top D_g^{-1} x_j + (\mathcal{T}_g^\top x_j)^\top \mathcal{S}^{-1} (\mathcal{T}_g^\top x_j). \quad (8.14)$$

If $D_g = L_g L_g^\top$ is a Cholesky factorization, its first term is $\|L_g^{-1} x_j\|^2$. Analogous triangular solves evaluate the second. An $N \times N$ map covariance is not needed for N standard errors. No arbitrary eigenvalue clipping or diagonal perturbation is hidden in these formulas.

8.6 Why the Schur complement is profile curvature

For a smooth local spatial optimum $\beta_*(\gamma)$, $L_\beta(\beta_*(\gamma), \gamma) = 0$. Differentiate this equation with respect to γ :

$$D \frac{\partial \beta_*}{\partial \gamma^\top} + F = 0, \quad \frac{\partial \beta_*}{\partial \gamma^\top} = -D^{-1}F.$$

For the profiled negative log likelihood $L_{\text{prof}}(\gamma) = L(\beta_*(\gamma), \gamma)$, the chain rule gives $\nabla_\gamma L_{\text{prof}} = L_\gamma$ because $L_\beta = 0$. Differentiating once more yields

$$\nabla_\gamma^2 L_{\text{prof}} = E + F^\top \frac{\partial \beta_*}{\partial \gamma^\top} = E - F^\top D^{-1}F = \mathcal{S}.$$

This identity concerns the observed Hessian along a differentiable interior solution branch with nonsingular D . Using expected information gives the corresponding efficient-information construction, not an exact data-specific profile derivative.

8.7 A closed-form Poisson nuisance adjustment

For the unpenalized full Poisson model, suppose the identified spatial basis contains a constant map: $Xc_B = \mathbf{1}_N$. Keep the spatial intercept available in each group and exclude confounded constant study columns so the full model is identifiable. Recall the scalar T_g , vector $t_g^{(z)}$, and matrix V_g defined in Section 3. Then

$$B_g c_B = X^\top \text{diag}(\mu_g^X) X c_B = X^\top \mu_g^X = a_g, \quad a_g^\top c_B = h_g.$$

Consequently, whenever $D_g = T_g B_g$ is invertible,

$$\mathcal{T}_g = D_g^{-1} F_g = \frac{c_B (t_g^{(z)})^\top}{T_g}, \quad (8.15)$$

$$F_g^\top D_g^{-1} F_g = \frac{h_g t_g^{(z)} (t_g^{(z)})^\top}{T_g}, \quad (8.16)$$

$$\mathcal{S} = \sum_g h_g \left\{ V_g - \frac{t_g^{(z)} (t_g^{(z)})^\top}{T_g} \right\} = \sum_g h_g T_g \text{Cov}_{\pi_g}(Z_i^\top), \quad \pi_{ig} = \mu_i^Z / T_g. \quad (8.17)$$

The last equality follows by writing the weighted covariance as $\sum_i \pi_{ig} Z_i^\top Z_i - \bar{z}_g \bar{z}_g^\top$ with $\bar{z}_g = t_g^{(z)} / T_g$. Thus information on shared covariates after estimating group baselines is exactly weighted *within-group* variation in study covariates. For any direction a ,

$$a^\top \mathcal{S} a = \sum_g h_g T_g \sum_{i \in \mathcal{I}_g} \pi_{ig} \{Z_i a - \bar{z}_g^\top a\}^2.$$

It vanishes precisely when $Z_i a$ is constant within each group. This makes the identification condition explicit and computes the small Schur matrix without large spatial solves under the stated Poisson assumptions. It is not a replacement for the general NB, penalized, or constrained formulas. At an interior Poisson optimum, the spatial intercept score additionally gives $h_g T_g = Y_g$; then (8.17) agrees with the allocation information derived after Equation (3.15).

9 Roughness penalties and uncertainty interpretations

9.1 A declared quadratic convention, not an invented operator

Paper Section 2.1.1 introduces a $P \times P$ roughness matrix J and refers its construction to Supplementary Appendix A.1. That supplement is not supplied within the reference PDF used here. We therefore do not assert an exact derivative operator, integration domain, boundary condition, or normalization for its J . The following generic calculation applies to a supplied fixed symmetric $J \succeq 0$ and declared nonnegative smoothing strengths τ_g :

$$\mathcal{P}(\vartheta) = \frac{1}{2} \sum_g \tau_g \beta_g^\top J \beta_g = \frac{1}{2} \vartheta^\top K \vartheta, \quad K = \text{blkdiag}(\tau_1 J, \dots, \tau_G J, 0_{R \times R}). \quad (9.1)$$

This explicitly chooses the factor $1/2$. If a convention omits that factor, the same numerical τ_g produces twice the gradient and twice the curvature.

Expand one term as $\tau_g \sum_{ab} \beta_{ga} J_{ab} \beta_{gb} / 2$. Differentiating with respect to β_{gc} gives

$$\partial_{\beta_{gc}} \mathcal{P} = \frac{\tau_g}{2} \left(\sum_b J_{cb} \beta_{gb} + \sum_a \beta_{ga} J_{ac} \right) = \tau_g (J \beta_g)_c, \quad (9.2)$$

$$\partial_{\beta_{gc} \beta_{ge}}^2 \mathcal{P} = \tau_g J_{ce}. \quad (9.3)$$

Thus for $L_{\text{pen}} = -\ell + \mathcal{P}$,

$$\nabla L_{\text{pen}} = -U + K\vartheta, \quad H_{\text{pen}} = H + K. \quad (9.4)$$

Separate group penalties preserve the group blocks and shared border. The penalized score equation is $U(\hat{\vartheta}) = K\hat{\vartheta}$, not $U(\hat{\vartheta}) = 0$.

9.2 Sampling covariance versus posterior inverse curvature

Expand the penalized equation near an unpenalized target ϑ_0 :

$$0 \approx U(\vartheta_0) - H(\vartheta_0)(\hat{\vartheta} - \vartheta_0) - K\vartheta_0 - K(\hat{\vartheta} - \vartheta_0), \quad (9.5)$$

$$(H + K)(\hat{\vartheta} - \vartheta_0) \approx U(\vartheta_0) - K\vartheta_0. \quad (9.6)$$

Under a correctly specified likelihood, replacing sensitivity by I gives the local approximations

$$\text{bias}(\hat{\vartheta}) \approx -(I + K)^{-1}K\vartheta_0, \quad V_{\text{freq}} \approx (I + K)^{-1}I(I + K)^{-1}. \quad (9.7)$$

Strong smoothing can change the population target; this linearization is not a global exact bias identity. By contrast a Gaussian-prior interpretation of K leads to a Laplace posterior covariance approximation $(H + K)^{-1}$. Only an actually quadratic posterior would make this exact. In expected-curvature notation the difference is

$$(I + K)^{-1} - (I + K)^{-1}I(I + K)^{-1} = (I + K)^{-1}K(I + K)^{-1} \succeq 0. \quad (9.8)$$

Thus inverse penalized curvature alone is not generally the frequentist sampling variance of a penalized estimator. Neither variance includes squared smoothing bias about the unpenalized target. For scalar information $i > 0$ and penalty k , the two variances are $1/(i + k)$ and $i/(i + k)^2$, respectively.

9.3 Smoothing selection and identifiability remain separate

All preceding derivatives condition on fixed τ_g . If a data-driven rule selects them, differentiating the score equation along a smooth solution branch gives

$$(H + K) \frac{\partial \hat{\vartheta}}{\partial \tau_g} = -\frac{\partial K}{\partial \tau_g} \hat{\vartheta}. \quad (9.9)$$

A joint delta calculation must include uncertainty in $\hat{\tau}$ and its covariance with the regression fit, or a bootstrap must repeat the selection rule. Merely adding an independent variance term is unjustified in general. For $H, K \succeq 0$, $H + K \succ 0$ precisely when $\ker(H) \cap \ker(K) = \{0\}$, because both quadratic forms are nonnegative. A roughness penalty leaving constant maps unpenalized cannot resolve the constant-map confounding direction in (1.7). Adding a numerical ridge changes the inverse and possibly the estimator; it is not evidence that the original information supported reliable tests.

10 Group maps, covariate contrasts, and Wald inference

10.1 Linear map evaluation and full voxelwise group covariance

The baseline log-intensity map is $\eta_g^X = X\beta_g$, an N -vector. For its coefficient covariance V_{gg} , the dimensionally correct identity is

$$\text{Var}(\hat{\eta}_g^X) \approx XV_{gg}X^\top, \quad (10.1)$$

not $X^\top V_{gg} X$ when X is $N \times P$. For all groups at voxel j , define the auxiliary evaluation matrix

$$A_j = (I_G \otimes x_j^\top, 0_{G \times R}). \quad (10.2)$$

Let $g_1, \dots, g_S$ be the S groups involved in the test and let $P_G \in \mathbb{R}^{S \times G}$ select them in that order. On the log-intensity scale recommended by the paper,

$$\theta_j = \begin{pmatrix} \log \mu_{g_1 j}^X \\ \vdots \\ \log \mu_{g_S j}^X \end{pmatrix} = P_G A_j \vartheta, \quad V_j = P_G A_j V A_j^\top P_G^\top. \quad (10.3)$$

Here θ_j and V_j now have exactly their paper Equation 12 meanings and dimensions. V is the chosen regression covariance estimate, so V_j is its induced voxel covariance estimate. Its (a, b) entry is $x_j^\top V_{\beta_{g_a} \beta_{g_b}} x_j$, including the cross-group terms induced by estimating shared γ . For the paper's contrast $C \in \mathbb{R}^{m \times S}$, define a new *coefficient-space* matrix $\mathcal{C}_j = C P_G A_j$. Then

$$C \theta_j = \mathcal{C}_j \vartheta, \quad C V_j C^\top = \mathcal{C}_j V \mathcal{C}_j^\top. \quad (10.4)$$

The covariance can therefore be computed by selected coefficient solves without forming V or any $N \times N$ map covariance. This evaluates the spatial baseline at the zero covariate row. For a prediction at fixed covariate row $Z_* \in \mathbb{R}^{1 \times R}$, append Z_* to the appropriate spatial evaluation row. For group contrasts whose row sums are zero, the common $Z_* \gamma$ term cancels; it does not cancel for arbitrary contrasts.

10.2 Wald rank, degrees of freedom, and estimability

For the paper's null $H_0 : C \theta_j = 0$, put $d_j = C \hat{\theta}_j$ and $\Omega_j = C V_j C^\top$. With m linearly independent estimable rows and $\Omega_j \succ 0$,

$$Q_j = (C \hat{\theta}_j)^\top (C V_j C^\top)^{-1} (C \hat{\theta}_j) \stackrel{H_0}{\approx} \chi_m^2. \quad (10.5)$$

This is paper Equation 12; Q_j is only a helper name for its squared statistic. To see the degrees of freedom, take $\Omega_j = L_j L_j^\top$. Under the null, asymptotic normality gives $L_j^{-1} d_j \approx N_m(0, I_m)$, and $Q_j = \|L_j^{-1} d_j\|^2$ is approximately a sum of m independent squared standard normals. For a nonzero specified null $C \theta_j = d_0$, replace d_j by $C \hat{\theta}_j - d_0$. The approximation requires asymptotic normality and an appropriate covariance estimate, not merely an invertible numerical matrix. Redundant rows must be removed, or a generalized-inverse version uses the rank of the estimable covariance on its supported null space. Unidentified directions cannot be made testable by a pseudoinverse alone. With a singular design, an estimable contrast must annihilate its null directions under the chosen constraints.

For one scalar contrast with positive standard error,

$$W_j = \frac{C \hat{\theta}_j}{\sqrt{C V_j C^\top}} \stackrel{H_0}{\approx} N(0, 1), \quad Q_j = W_j^2, \quad C \in \mathbb{R}^{1 \times S}. \quad (10.6)$$

This preserves the signed W_j of paper Equation 13, instead of renaming it Z and colliding with the study design. For example, take $S = 2$, $C = (1, -1)$, and groups g, h . Then

$$W_j = \frac{x_j^\top (\hat{\beta}_g - \hat{\beta}_h)}{\sqrt{x_j^\top (V_{\beta_g \beta_g} + V_{\beta_h \beta_h} - V_{\beta_g \beta_h} - V_{\beta_h \beta_g}) x_j}}.$$

The two cross-covariances follow directly by expanding the variance of a difference, rather than assuming independent group estimates. No universal “sum-to-one” condition is imposed on a contrast. Differences such as $(1, -1)$ sum to zero, averages can sum to one, and coefficient tests use other valid rows. The null determines the rows; independence and estimability determine the degrees of freedom.

10.3 Shared global-covariate inference

Select s of the R global covariates with a row-selection matrix $P_\gamma \in \mathbb{R}^{s \times R}$ and define $\gamma_{\text{sel}} = P_\gamma \gamma$. The paper's C_γ has dimension $m \times s$, not $m \times R$ unless all covariates are selected. The selection and contrast steps give

$$V_{\text{sel}} = P_\gamma V_{\gamma\gamma} P_\gamma^\top, \quad \text{Var}(C_\gamma \hat{\gamma}_{\text{sel}}) \approx C_\gamma V_{\text{sel}} C_\gamma^\top.$$

For $H_0 : C_\gamma \gamma_{\text{sel}} = 0$, paper Equation 14 is therefore

$$Q_\gamma = (C_\gamma \hat{\gamma}_{\text{sel}})^\top (C_\gamma V_{\text{sel}} C_\gamma^\top)^{-1} (C_\gamma \hat{\gamma}_{\text{sel}}) \approx \chi_m^2. \quad (10.7)$$

Under the fixed-dispersion inverse-information convention, $V_{\text{sel}} = P_\gamma \mathcal{S}^{-1} P_\gamma^\top$. Joint dispersion adjustment, robust covariance, or penalized sampling covariance can require the corresponding replacement. All m rows must again be independent and estimable. For a specified nonzero null, subtract its m -vector on both sides of the quadratic form. The conditional inverse E^{-1} is not generally $V_{\gamma\gamma}$. Even a small R does not guarantee stable covariate inference: $\mathcal{S}$ can be nearly singular when covariates are confounded with group baselines or with one another. Nonzero fitted coefficients do not prove information rank. Low dimension helps computation, not identification. Because covariates have units and preprocessing scales, comparisons between two components of γ must specify that scale.

10.4 Exponentiation and optional normalization

For the baseline intensity $\mu_g^X = \exp(X\beta_g)$, componentwise differentiation gives $d\mu_g^X = \text{diag}(\mu_g^X) X d\beta_g$. Therefore the first-order delta covariance is

$$\text{Cov}(\hat{\mu}_g^X, \hat{\mu}_h^X) \approx \text{diag}(\mu_g^X) X V_{\beta_g \beta_h} X^\top \text{diag}(\mu_h^X). \quad (10.8)$$

At a voxel, $\text{Var}(\hat{\mu}_{gj}^X) \approx (\mu_{gj}^X)^2 \text{Var}(x_j^\top \hat{\beta}_g)$. The map from β_g to log intensity is linear; the exponentiation step introduces an additional delta approximation. If the paper's θ_j is instead taken on the intensity scale, replace (10.3) by $\theta_j = \exp(P_G A_j \vartheta)$ and use Jacobian $\text{diag}(\theta_j) P_G A_j$ to form V_j . Using the log-map covariance unchanged in an intensity-scale test would omit these mean factors. A log-scale interval exponentiates to a positive intensity interval, and a log-difference interval exponentiates to an intensity-ratio interval.

As an optional derived output, normalize $q_g = \mu_g^X / h_g$. Since $dh_g = (\mu_g^X)^\top X d\beta_g$, the quotient rule gives

$$\frac{\partial q_g}{\partial \beta_g^\top} = \{\text{diag}(q_g) - q_g q_g^\top\} X. \quad (10.9)$$

The row derivatives sum to zero because $\sum_j q_{gj} = 1$. Its covariance is necessarily singular in that normalization direction. This optional probability map is not the same estimand as the paper's unnormalized baseline intensity; normalize only if that is the intended scientific question.

10.5 Homogeneous references estimated from the data

For a genuinely known homogeneous log baseline η_{g0} , the scalar null $x_j^\top \beta_g = \eta_{g0}$ uses the standard contrast variance. If η_{g0} is estimated from the same data, instead

$$\text{Var}(x_j^\top \hat{\beta}_g - \hat{\eta}_{g0}) = \text{Var}(x_j^\top \hat{\beta}_g) + \text{Var}(\hat{\eta}_{g0}) - 2 \text{Cov}(x_j^\top \hat{\beta}_g, \hat{\eta}_{g0}). \quad (10.10)$$

The estimated reference is not automatically a fixed null constant. For example, a linear reference $a^\top \beta_g$ is tested with row $(x_j - a)^\top$, not with $x_j^\top$ alone. A separately constrained homogeneous fit may require joint influence calculations or a null bootstrap repeating the estimation. For a concrete alternative reference derived from the fitted map, take $\eta_{g0} = \log(h_g/N)$, the log of

its mean intensity. This is not asserted to be the same as a separately refitted null. Because $dh_g = a_g^\top d\beta_g$,

$$\nabla_{\beta_g} \{x_j^\top \beta_g - \log(h_g/N)\} = x_j - a_g/h_g,$$

so its local delta variance is

$$(x_j - a_g/h_g)^\top V_{\beta_g \beta_g} (x_j - a_g/h_g),$$

evaluated at the fitted parameters. The centered log intensity is $\log(Nq_{gj})$, which also exhibits its relation to the optional normalized map. The same qualification applies after exponentiating the reference. Finally, voxelwise p -values are not already familywise-corrected results; multiple-testing procedures require their own assumptions and a clearly specified testing family.

11 Resampling and an optional robust-covariance extension

11.1 Study sandwich: an extension, not the core paper calculation

For a genuine study-factorized likelihood, let $U_i = \nabla_{\vartheta} \ell_i$ and $H = -\sum_i \nabla_{\vartheta} U_i^\top$. A standard study-cluster sandwich estimate is

$$\hat{V}_{\text{rob}} = H^{-1} \left(\sum_i U_i U_i^\top \right) H^{-1}. \quad (11.1)$$

These are sums, so no additional $1/M$ factor is inserted. Its usual justification uses independent studies, enough independent units, identification, and appropriate estimating-equation regularity. Poisson uses $U_i = \sum_j b_{ij}(Y_{ij} - \mu_{ij})$. The underlying independent NB uses its weighted residual score, whereas clustered NB uses $U_i = B_i^\top (Y_i - c_i \mu_i)$. Substituting Poisson residuals into a fitted NB sandwich changes the estimating equation and is not a general NB robust covariance.

The aggregate NB likelihood is a function of group-voxel totals and nonlinear moment maps across studies. It does not provide arbitrary study-factorized likelihood scores. A valid influence calculation for that aggregate estimator, or refitting after study resampling, requires separate justification. Assigning each study a convenient fraction of an aggregate score does not establish (11.1).

For selected coefficient contrasts solve $HB = \mathcal{C}^\top$ and accumulate

$$\mathcal{C} \hat{V}_{\text{rob}} \mathcal{C}^\top = \sum_i (B^\top U_i)(B^\top U_i)^\top. \quad (11.2)$$

This is a PSD Gram matrix with rank at most $\min(M, k, GP + R)$. At an exactly solved unpenalized score equation, $\sum_i U_i = 0$ reduces the bound to $M - 1$. Few studies can therefore limit inference even with many observed foci. Joint nuisance estimation requires corresponding joint scores and bread. For fixed penalties the bread is $H + K$, while the deterministic penalty is not an extra independent random score in the meat. Score centering and the penalized population target require attention under substantial shrinkage; a naive uncentered meat is not universal.

11.2 Null resampling tailored to the tested hypothesis

Paper Section 2.1.4 describes randomization under spatial homogeneity using a binomial process and group comparison using a pooled shared intensity fit followed by regeneration and refitting. These procedures simulate a specified null; they are not distribution-free. A careful operational specification is:

1. State the null, including which group maps coincide or are homogeneous, whether study totals are conditioned on, the voxel mask, covariates, dispersion convention, and the count-generating distribution.
2. Fit the null, estimating its nuisance parameters under its constraints. For shared-baseline group comparisons retain the intended covariate design and the numbers of studies in each group.
3. Generate independent studies under that null. For clustered NB, draw one gamma multiplier per study and then its conditionally independent counts; drawing one new multiplier per voxel changes the model.
4. Apply the same preprocessing conventions and refit the same alternative or statistic-producing procedure used on the observed data. Re-estimate fitted dispersion, and repeat penalty selection if it was part of the original procedure rather than a prespecified constant.
5. Compare the observed statistic with the simulated null statistics, using the chosen one-sided, two-sided, or maximum-statistic rule.

For aggregate NB, distinguish simulating the matched totals distribution from simulating original independent NB studies and then aggregating. They assess different distributions except in an exact closure case. If the statistic requires study totals or study-level resampling, aggregate totals alone do not uniquely specify a study allocation.

11.3 Replicates, failed fits, and what the foci threshold means

For B successful null replicates with statistics T_b , a common Monte Carlo tail estimate is

$$\hat{p} = \frac{1 + \sum_{b=1}^B \mathbf{1}_{\{T_b \geq T_{\text{obs}}\}}}{B + 1}. \quad (11.3)$$

The comparison statistic must encode sidedness consistently. The smallest attainable value is $1/(B+1)$, which matters for multiple-testing resolution. Conditional on an ideal fixed null, Monte Carlo variation is on the order of $\sqrt{p(1-p)/B}$; nuisance estimation adds other approximation error. The paper describes at least 1,000 repetitions. This is a reported procedural choice, not a universal guarantee of accurate extreme tails.

Predefine convergence checks and a failed-fit policy. Record attempted and successful fits, retry rules, boundary estimates, and sensitivity to failures. Silently deleting difficult fits can select a nonrepresentative null distribution. Resampling whole independent studies, typically within groups for group-specific sampling targets, preserves their within-study dependence. Voxelwise resampling does not. A nonparametric study bootstrap and a parametric null bootstrap address different questions; a hypothesis test needs an appropriate null mechanism.

The paper’s practical criterion of approximately 200 foci for reliable inverse-information inference is empirical guidance from its setting. It is not a theorem about rank, coverage, or every design. Rank and conditioning also depend on spline support, study allocation, covariate variation, dependence, and fitted intensity. Bootstrap avoids directly using some unstable inverses but cannot automatically cure nonidentification, poor model fit, or failed optimization.

12 Numerical audit, conditioning, and computational accounting

12.1 Symmetric eigenvalues and scale-aware diagnostics

For a symmetric positive-definite information matrix, the spectral condition number is

$$\kappa_2(H) = \lambda_{\max}(H)/\lambda_{\min}(H). \quad (12.1)$$

To derive this without a general singular-value decomposition, let $H = O\Lambda O^\top$ with O orthogonal. Then $H^\top H = O\Lambda^2 O^\top$, so the singular values are $|\lambda_a|$. For any symmetric nonsingular H ,

$$\kappa_2(H) = \frac{\max_a |\lambda_a|}{\min_a |\lambda_a|},$$

positive definiteness permits dropping the absolute values. A symmetric eigensolver therefore targets the same spectral condition number as a general SVD; it does not merely estimate a different norm. Inspect signed symmetric eigenvalues, not just magnitudes: a negative eigenvalue indicates indefinite curvature, whereas a tiny positive one indicates weak information or scaling. Check symmetry before applying a symmetric eigensolver; substantial antisymmetry indicates an assembly or differentiation problem. No universal threshold on κ_2 proves inferential validity.

Block conditioning alone is insufficient. For $H = \text{diag}(1, 10^{-8})$, each scalar block has condition number one, but the full matrix has condition number 10^8 . For independent blocks the characteristic polynomial is the product of their characteristic polynomials, so the full spectrum is their union. Use the global largest and smallest eigenvalue magnitudes. For a bordered matrix, Schur elimination is a congruence rather than an orthogonal similarity: the spectra of D and $\mathcal{S}$ do not generally combine into the spectrum of H . Border coupling can create a second problem:

$$H_\varepsilon = \begin{pmatrix} 1 & 1 - \varepsilon \\ 1 - \varepsilon & 1 \end{pmatrix}, \quad \lambda(H_\varepsilon) = \{\varepsilon, 2 - \varepsilon\}, \quad \mathcal{S} = 1 - (1 - \varepsilon)^2. \quad (12.2)$$

For small positive ε , all diagonal blocks are benign, yet the full covariance is unstable. This is near confounding through the shared border. Examine D_g , $\mathcal{S}$, and scale, not just blockwise rank.

If $\vartheta = A\phi$ for a fixed nonsingular scale matrix A , $H_\phi = A^\top H_\vartheta A$ and $V_\vartheta = AV_\phi A^\top$. Scaling can improve numerical solves while leaving correctly transformed contrasts unchanged. It cannot create information in a true null direction. For a computed solve $Hx = b$, inspect the relative residual $\|Hx - b\|/(\|H\|\|x\| + \|b\|)$; a small residual is a numerical check, not a proof that estimated standard errors have correct coverage.

12.2 Manual Schur example with two groups and one covariate

Take $G = 2$, $P = R = 1$, with

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad F = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad E = 2. \quad (12.3)$$

Then $\mathcal{T} = (1/2, 1/3)^\top$, $\mathcal{S} = 2 - 1/2 - 1/3 = 7/6$, and

$$H = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 3 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad H^{-1} = \frac{1}{7} \begin{pmatrix} 5 & 1 & -3 \\ 1 & 3 & -2 \\ -3 & -2 & 6 \end{pmatrix}. \quad (12.4)$$

Both D and $\mathcal{S}$ are positive definite, so (8.8) proves $H \succ 0$. Multiplying the displayed matrices gives identity. The fitted groups have covariance $1/7$, despite their zero direct information cross block. Their difference has variance $(5 + 3 - 2)/7 = 6/7$. Treating γ as fixed instead gives $1/2 + 1/3 = 5/6$. The shared-nuisance adjustment is $(1/2 - 1/3)^2(6/7) = 1/42$, so $5/6 + 1/42 = 6/7$. Adding the marginal joint variances while omitting covariance would give $8/7$, a different and incorrect result for this contrast. These are audit numbers, not a fitted example or a performance claim.

12.3 Manual moment-matching examples

For two studies take $\mu_1^Z = 1$, $\mu_2^Z = 2$, $\mu_j^X = 1/2$, and $\alpha = 1$. Then $s_1 = 3$, $s_2 = 5$, $s_3 = 9$, and

$$\mu_j = \frac{3}{2}, \quad \text{Var}(Y_j) = \frac{3}{2} + \frac{5}{4} = \frac{11}{4}, \quad r' = \frac{9}{5}, \quad A = \frac{3}{5}, \quad p'_j = \frac{5}{11}. \quad (12.5)$$

The matched variance is $\mu_j + \mu_j^2/r' = 3/2 + (9/4)/(9/5) = 11/4$, confirming the two-moment calculation. The third-cumulant discrepancy from (5.11) is $2(1/2)^3(9 - 25/3) = 1/6$, demonstrating nonexact closure.

For an exact common-probability case let $\mu_1^Z = \mu_2^Z = 1$ with the same $\mu_j^X = 1/2$ and $\alpha = 1$. Then $s_1 = s_2 = 2$, $r' = 2$, $A = 1$, $\mu_j = 1$, and $p'_j = 1/3$. Each original cell has shape one and probability $1/3$; their sum has shape two and the same probability. Its variance is $1 + 1/2 = 3/2$, equal to the sum of the original variances. These two examples separately check the approximation and the exact case.

12.4 Complexity of assembly, solves, and requested output

Let $d = GP + R$. The following are dense worst-case algebraic orders, not timings, speedups, or claims about a particular implementation. Spline sparsity, caching, and the number of requested contrasts can substantially change practical costs.

Operation	Dense order	Qualification
Full $d \times d$ factorization	$O(d^3)$	Storage $O(d^2)$, before any map output
Factor separate D_g	$O(GP^3)$	Storage $O(GP^2)$ for group factors
Solve $D_g \mathcal{T}_g = F_g$	$O(GP^2 R)$	Reuses the group factors
Form and factor $\mathcal{S}$	$O(GPR^2 + R^3)$	Border storage $O(GPR + R^2)$
k selected right-hand sides	$O(k[GP^2 + GPR + R^2])$	After factorization; no full inverse needed
Spatial weighted Gram assembly	$O(GNP^2)$	Reusable group terms for separable families
Study covariate moment assembly	$O(MR^2)$	First/second moments for Poisson and matched NB
All group map diagonals	$O(GNP^2 + GNPR + GNR^2)$	Dense triangular solves; chunk voxels to limit memory

Clustered projection also uses study-specific scalar weights and rank-one group corrections, as in (6.21). Aggregate NB global blocks use the inner R -dimensional moments and outer voxel sums, not an original independent-cell Hessian. The displayed assembly orders do not include fitting iterations, special-function costs, preprocessing, bootstrap replication, or the cost of constructing a roughness matrix. An efficient factorization changes neither the model nor the validity conditions of its covariance approximation.

12.5 Audit checklist and limits of this appendix

For manual or implementation review, check in this order:

1. Fix the paper's X, Z, β_g, γ dimensions and remove unidentified directions. State the observation model and data margins.
2. Fix original versus effective dispersion during each derivative. Keep factorial and gamma terms whenever their derivatives or absolute likelihood interpretation require them.
3. Verify scalar signs and every nonlinear inner-map term. For aggregate NB compare original and log-coordinate formulas; for clustered NB compare log-mean projection and total-mean differentiation.
4. Assemble D_g, F_g, E and compare selected full-system and Schur solves, including the signed symmetry and residual checks.

5. Declare observed or expected information, fixed or joint nuisance inference, and frequentist or posterior penalty interpretation.
6. Form estimable contrasts with the full required covariance and qualify the null, asymptotic reference, resampling law, and multiplicity rule.

All derivations are contained in this single source. The algebra was adapted from the repository's `cbmr_expanded_appendix.tex` and `expanded_covariance.tex` solely as provenance, not as required reading or generated inputs. No claim is made that every line has been verified in a proof assistant, that every optional extension is implemented, or that the unavailable original supplement has been reconstructed. Exact differentiation of a specified likelihood, exact block algebra, working-model accuracy, numerical reliability, and asymptotic coverage are distinct claims and must be assessed separately.

12.6 Editable source and independent build

The editable source is `docs/imag_a.1057_appendix.tex`. From the repository root:

```
make imag-appendix
```

The command `make paper-appendices` builds this document and the separate lesion-manuscript appendix. Neither command replaces the supplied papers or the combined expanded appendix.

References

- [1] Y. Yu, L. D. Hill-Bowen, M. C. Riedel, et al. CBMR: Coordinate-based meta-regression for group and covariate inference. *Imaging Neuroscience*, 3, 2025. DOI: [10.1162/IMAG.a.1057](https://doi.org/10.1162/IMAG.a.1057). Supplied reference: `docs/imag.a.1057.pdf`.